Comparing two things: T-tests

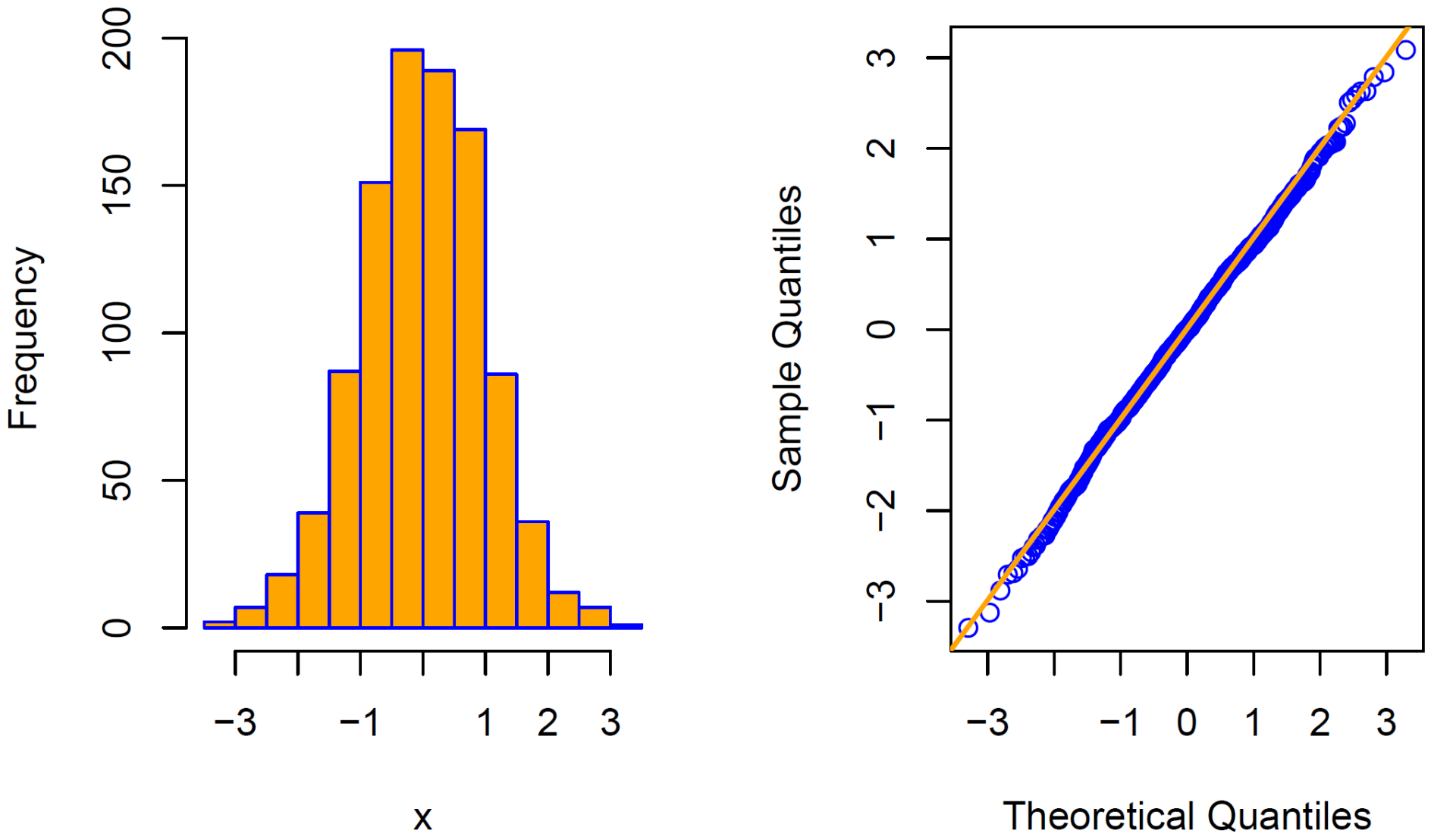
Rolando Coto-Solano. Dartmouth College

LING10, January 2026

(Something from last week: Q-Q Plots)

Q-Q plots show you the “quantiles” (“percentiles”) of your chart, plotted agains the percentiles of a theoretical normal distribution.

(Something from last week: Q-Q Plots)



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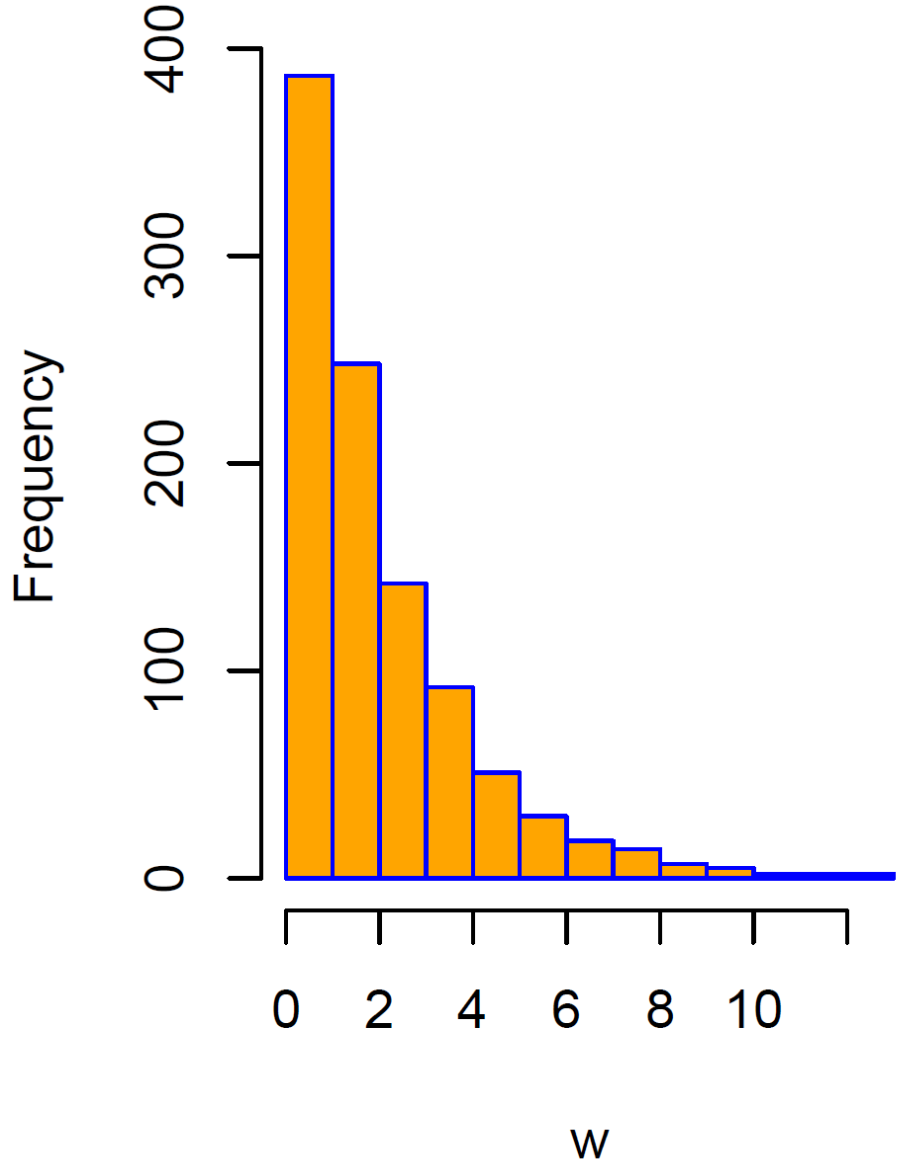

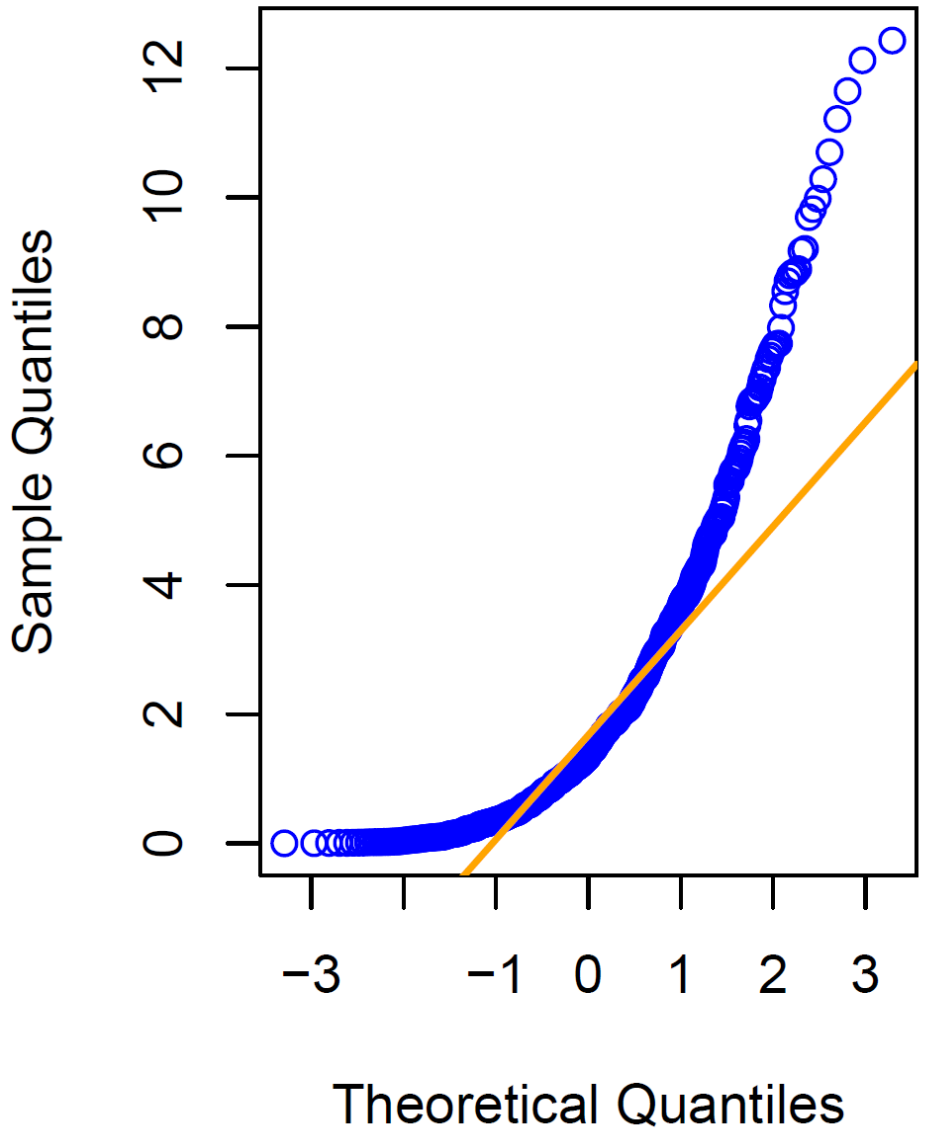

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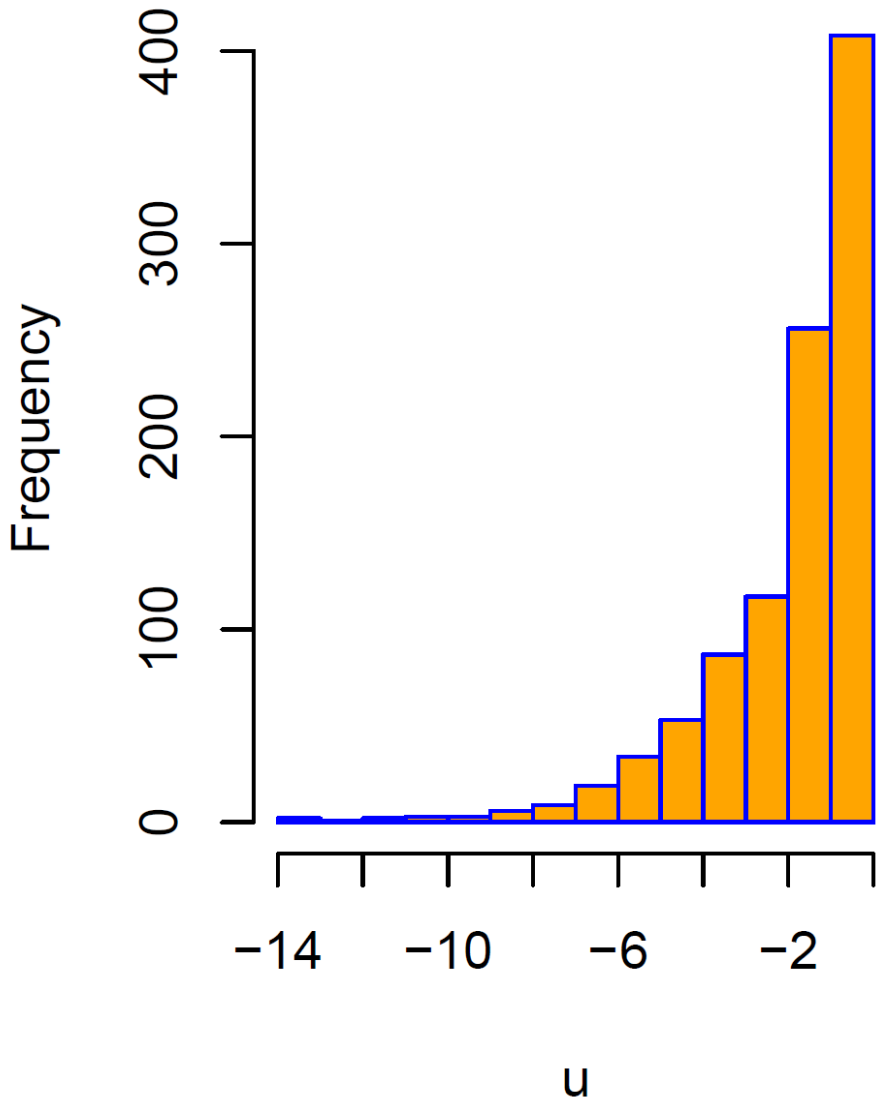

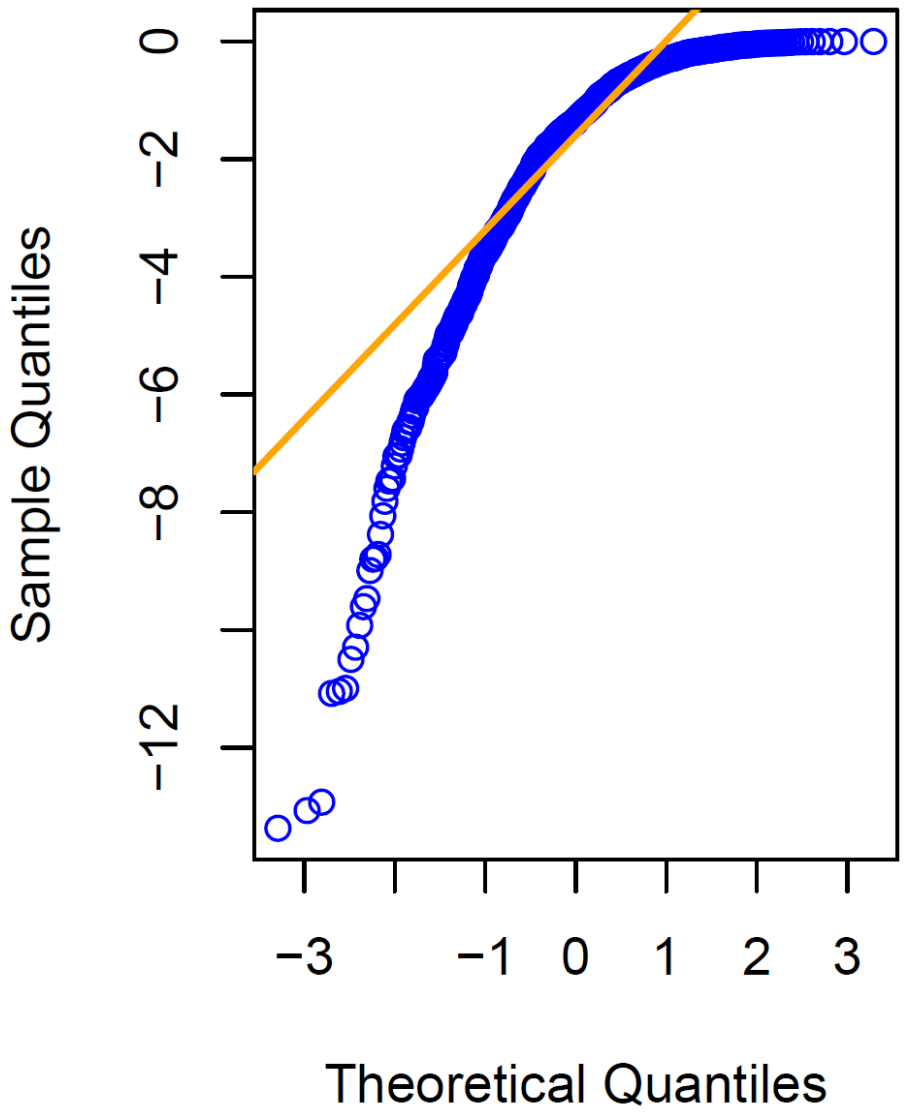

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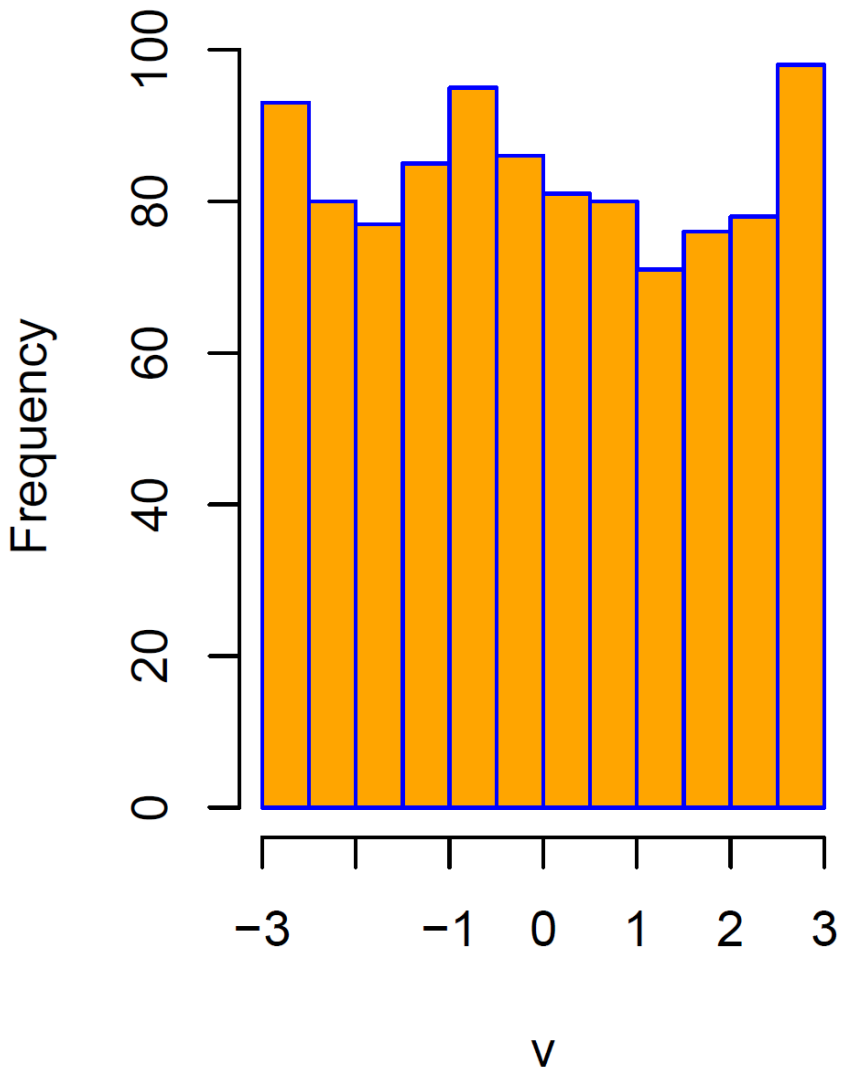

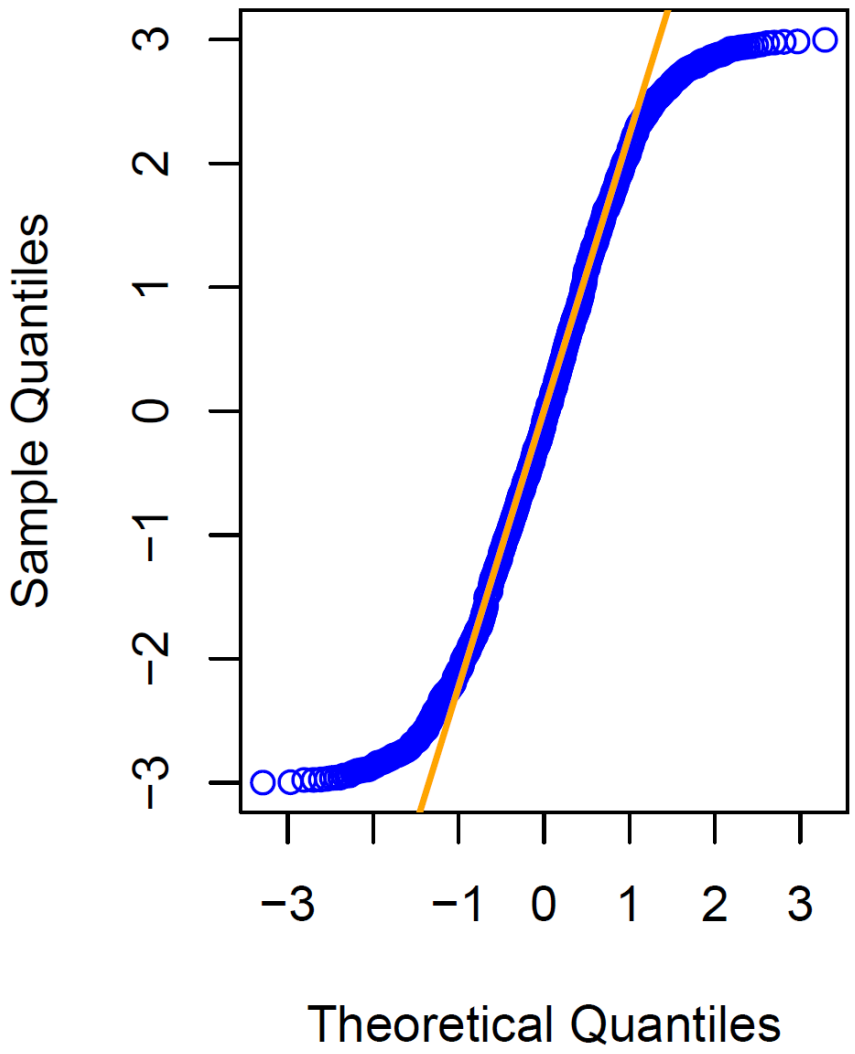

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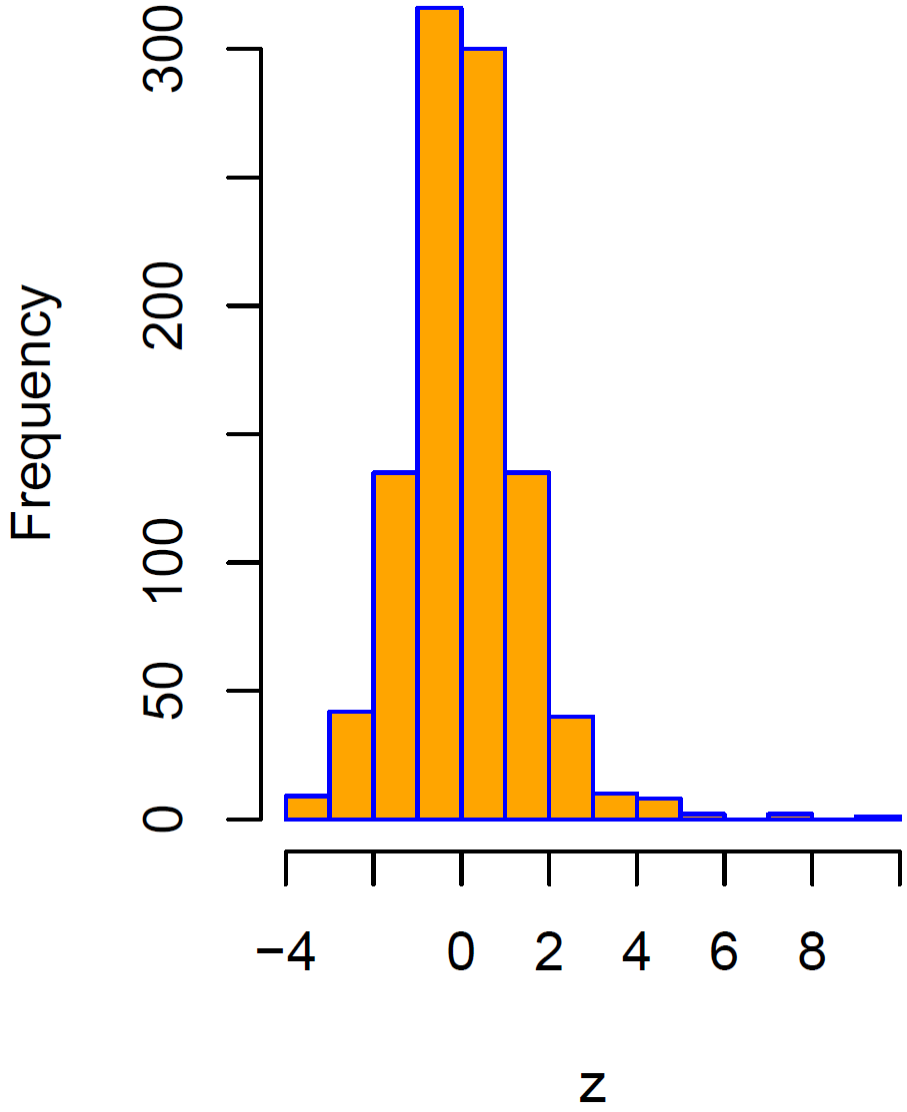

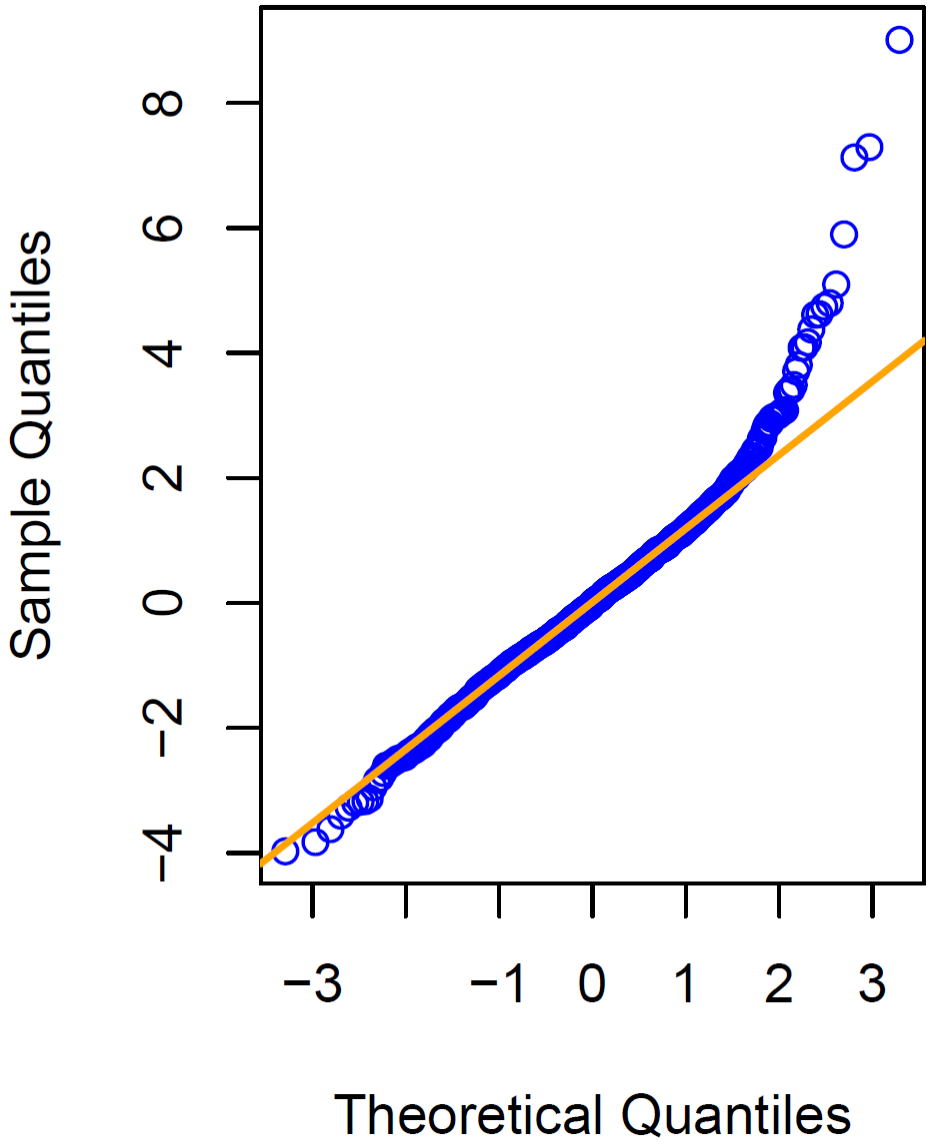

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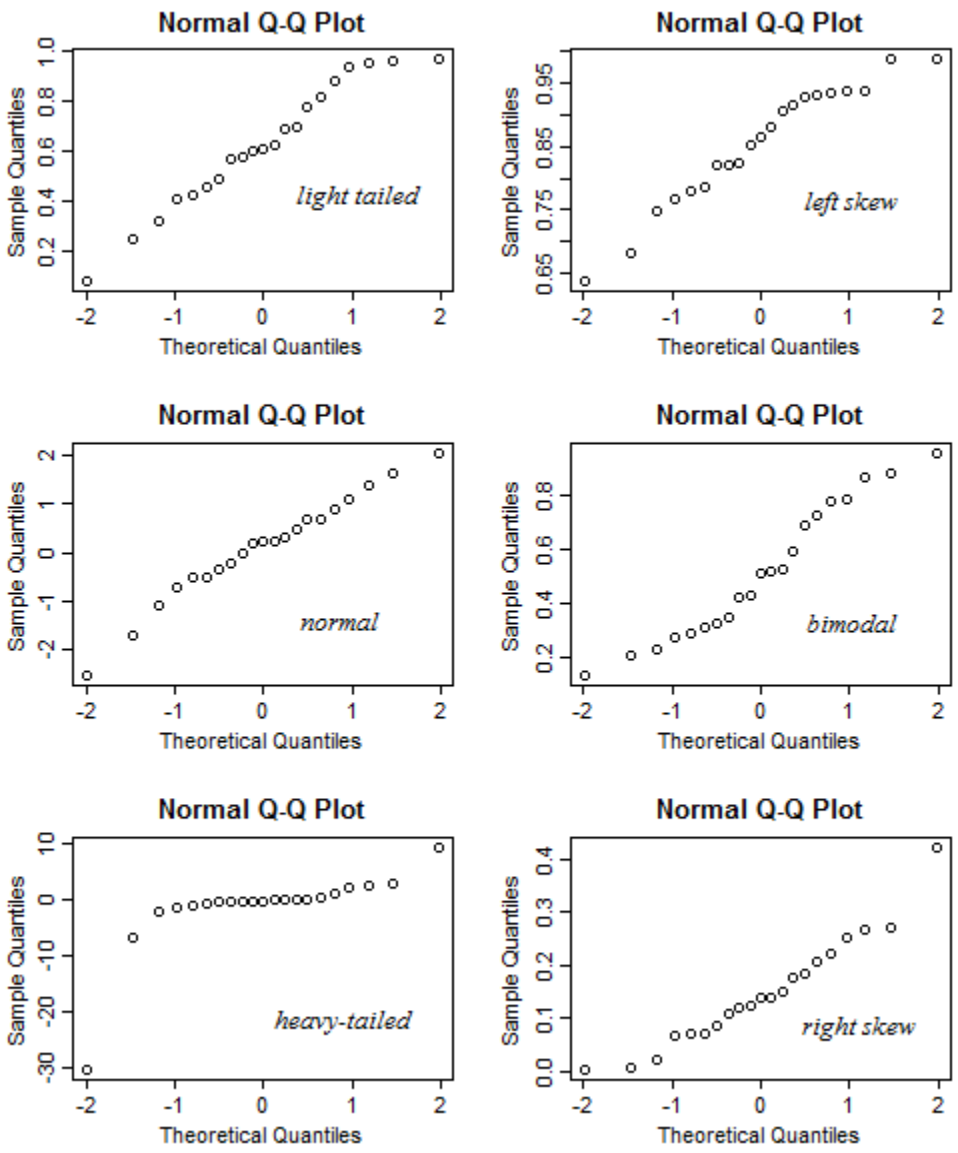
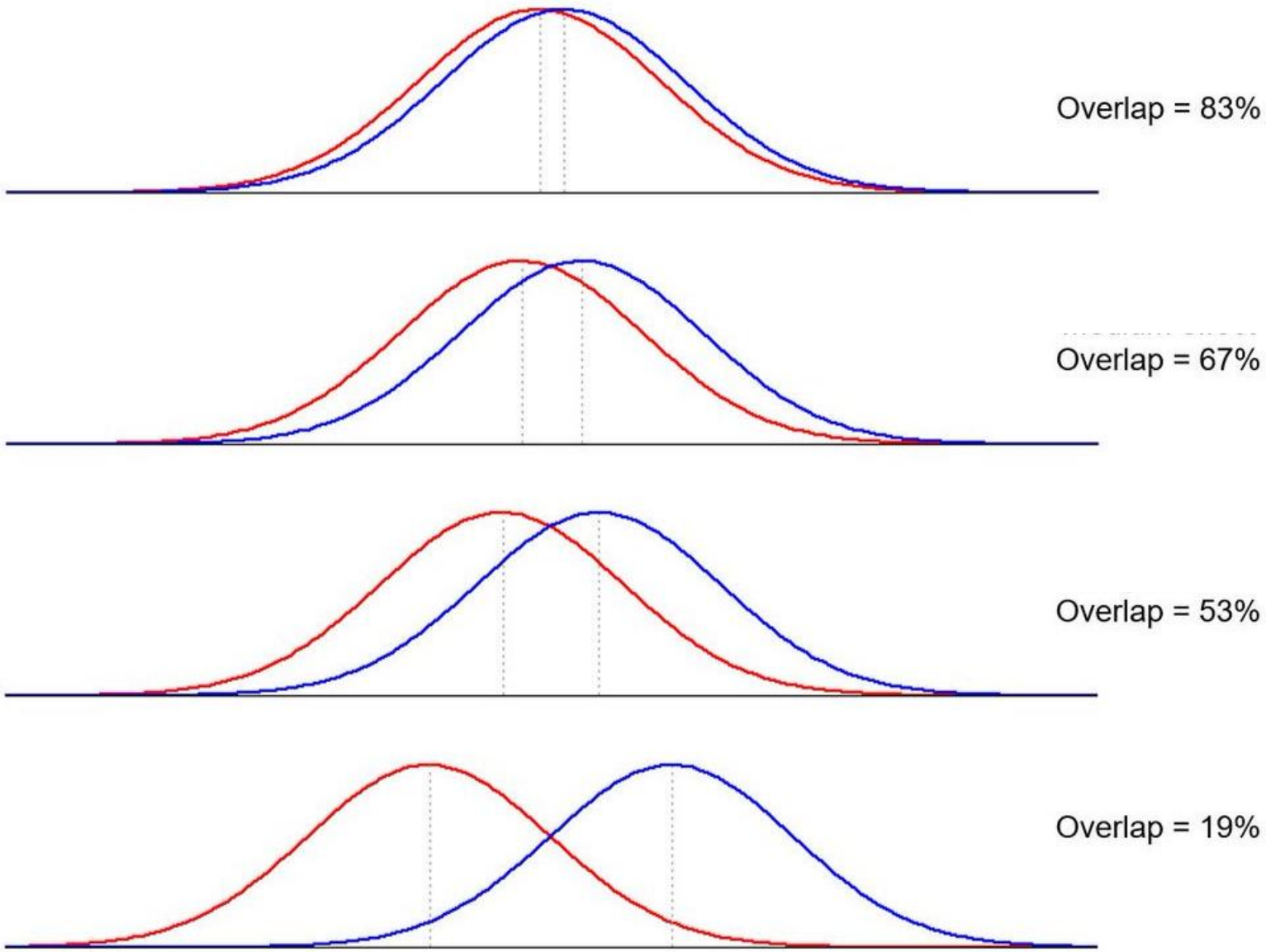


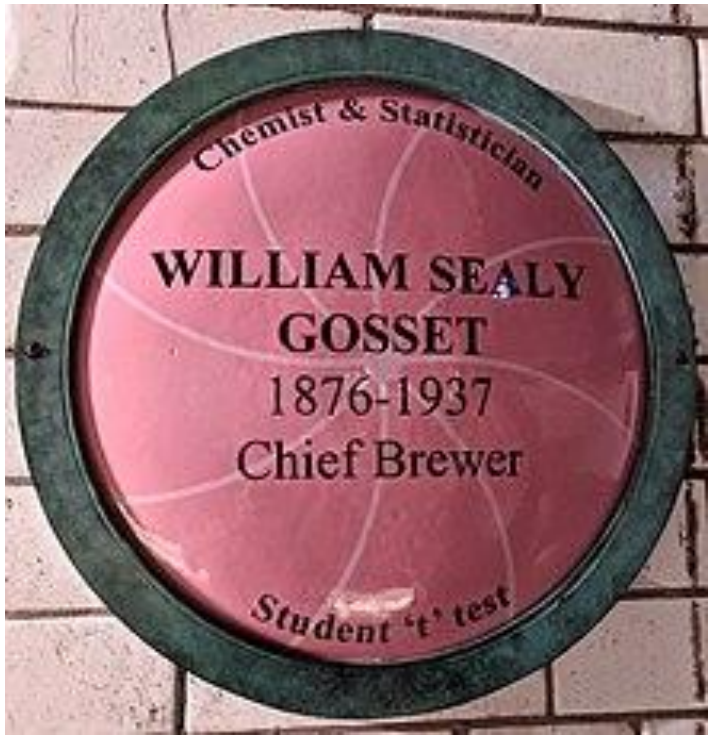
T-test of two independent samples

Are the means of these groups the same?

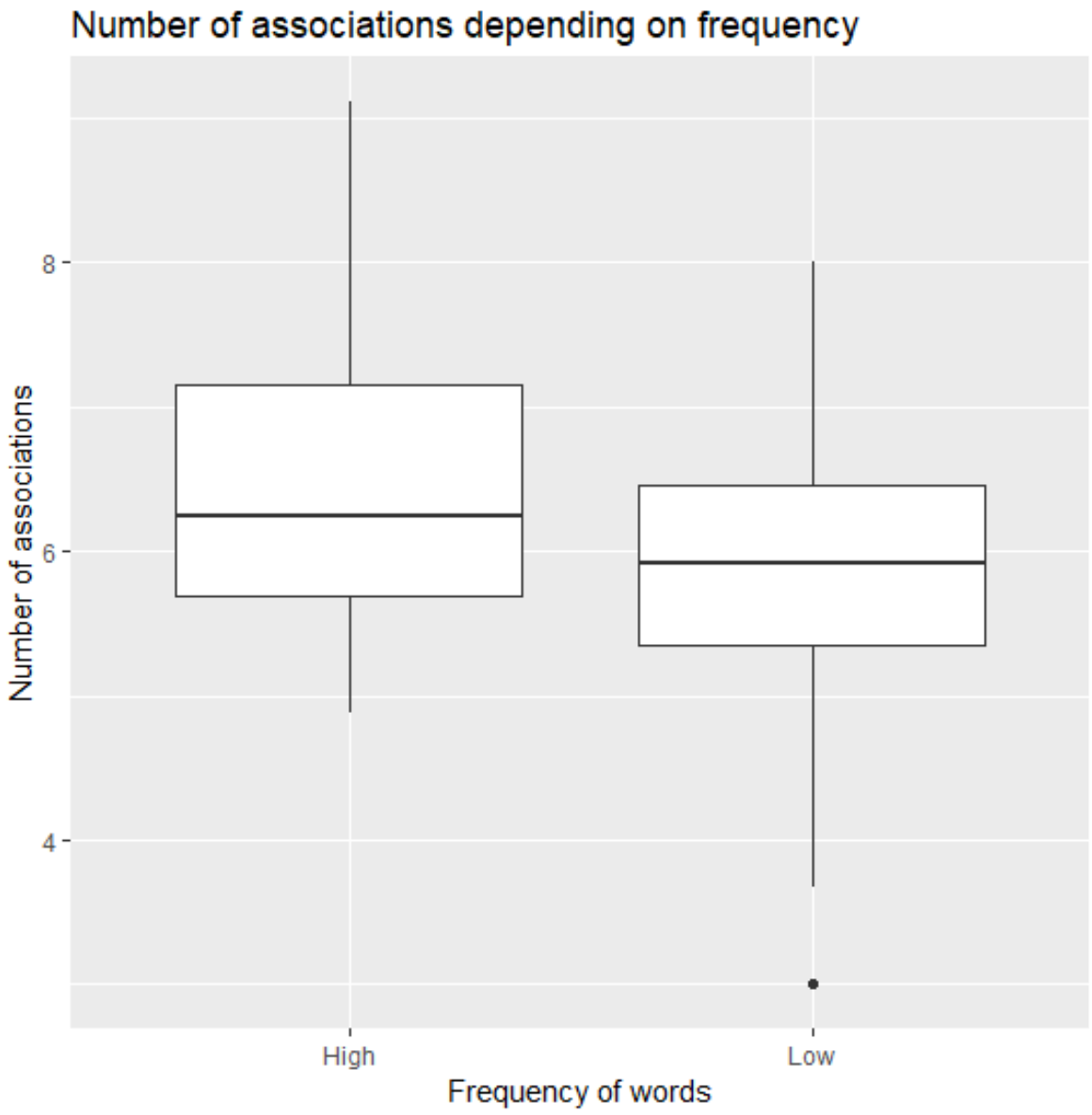


T-test of two independent samples

The first tool we have is the Student's **t-test**.



T-test of two independent samples

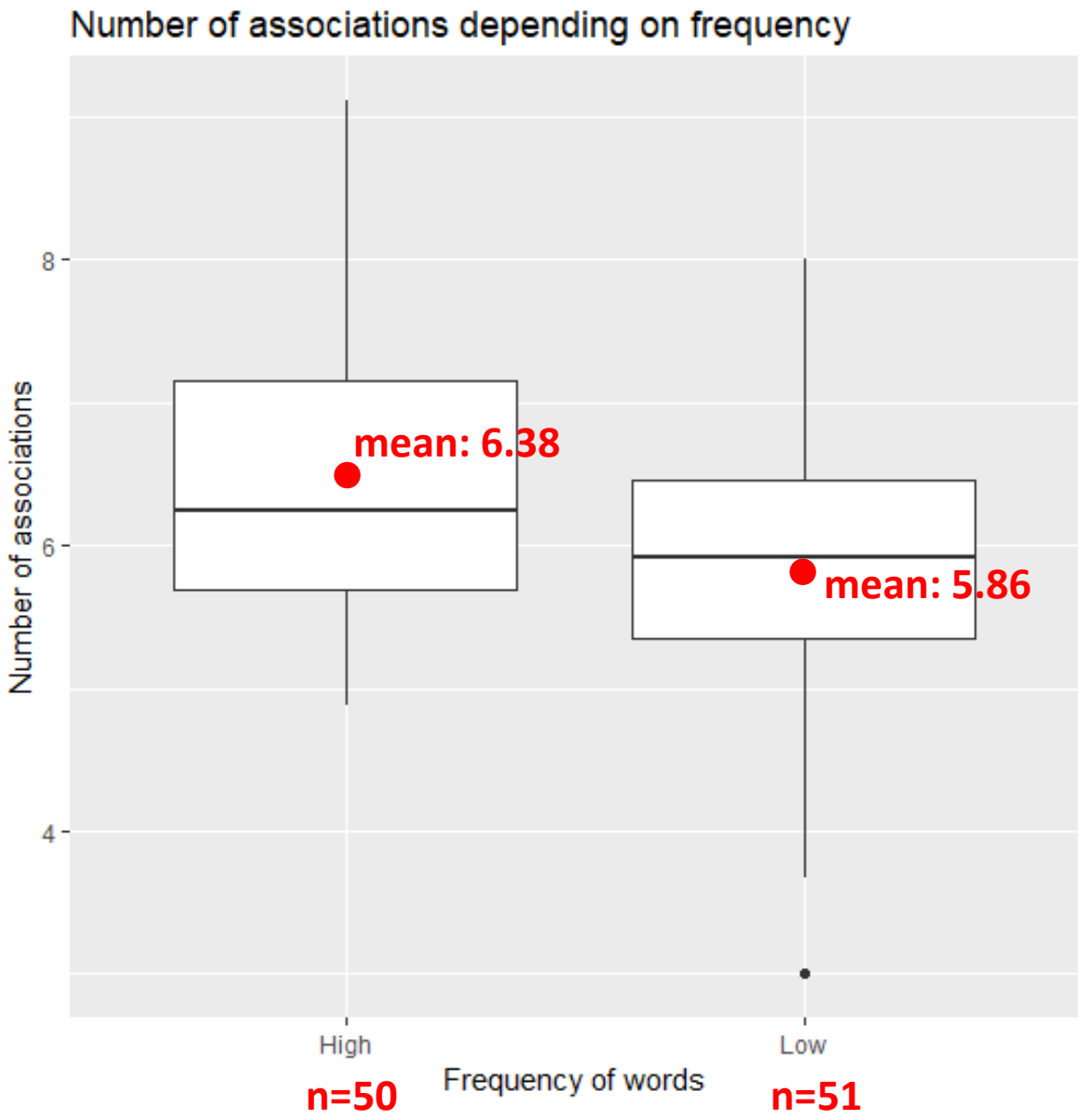


T-test of two independent samples

<u>High freq words:</u>

n = 50 words

mean = 6.38



<u>Low freq words:</u>

n = 51 words

mean = 5.86

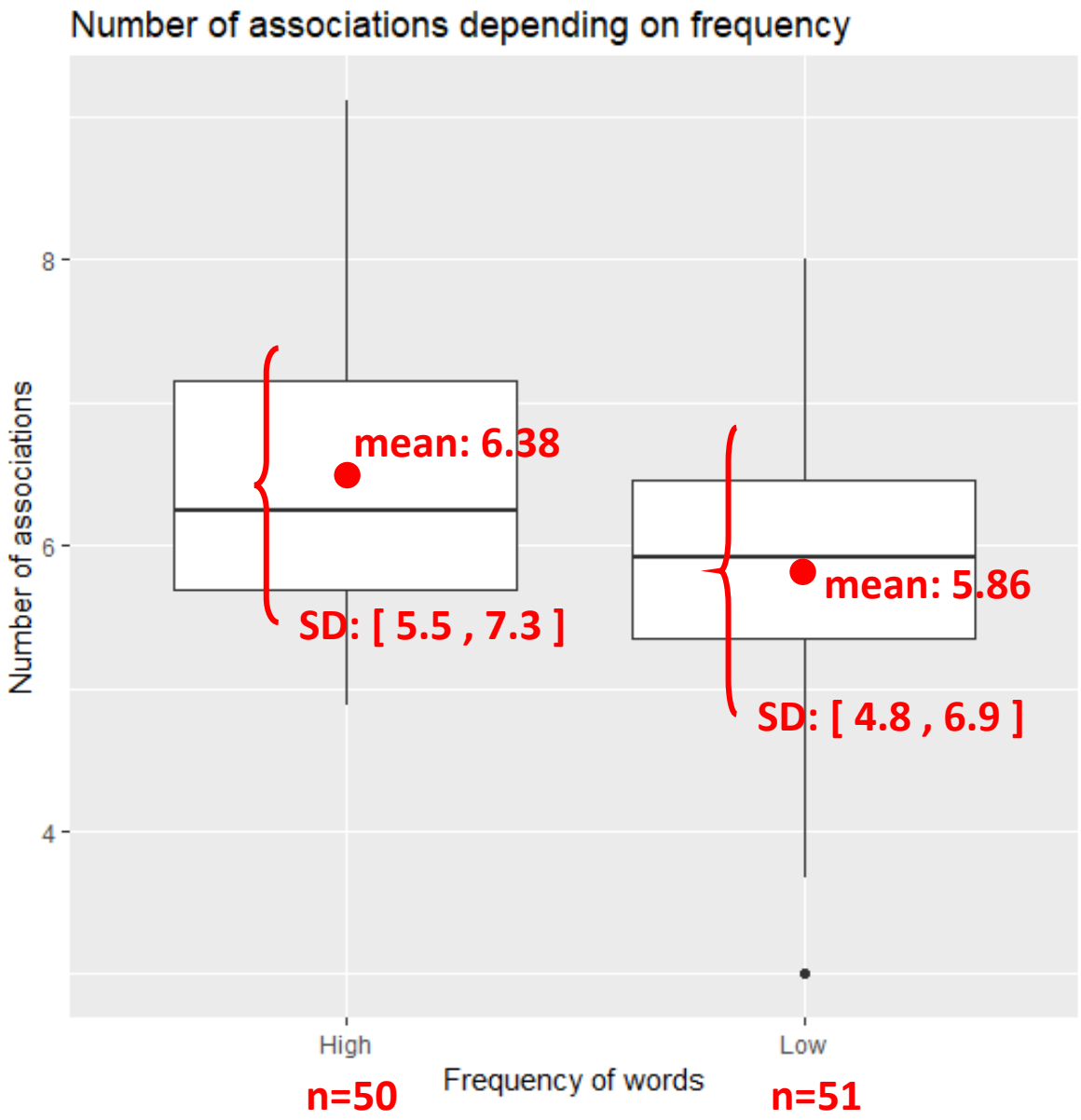
T-test of two independent samples

<u>High freq words:</u>

n = 50 words

mean = 6.38
SD = 0.9

mean + SD = 7.3
mean – SD = 5.5



<u>Low freq words:</u>

n = 51 words

mean = 5.86
SD = 1.03

mean + SD = 6.9
mean – SD = 4.8

T-test of two independent samples

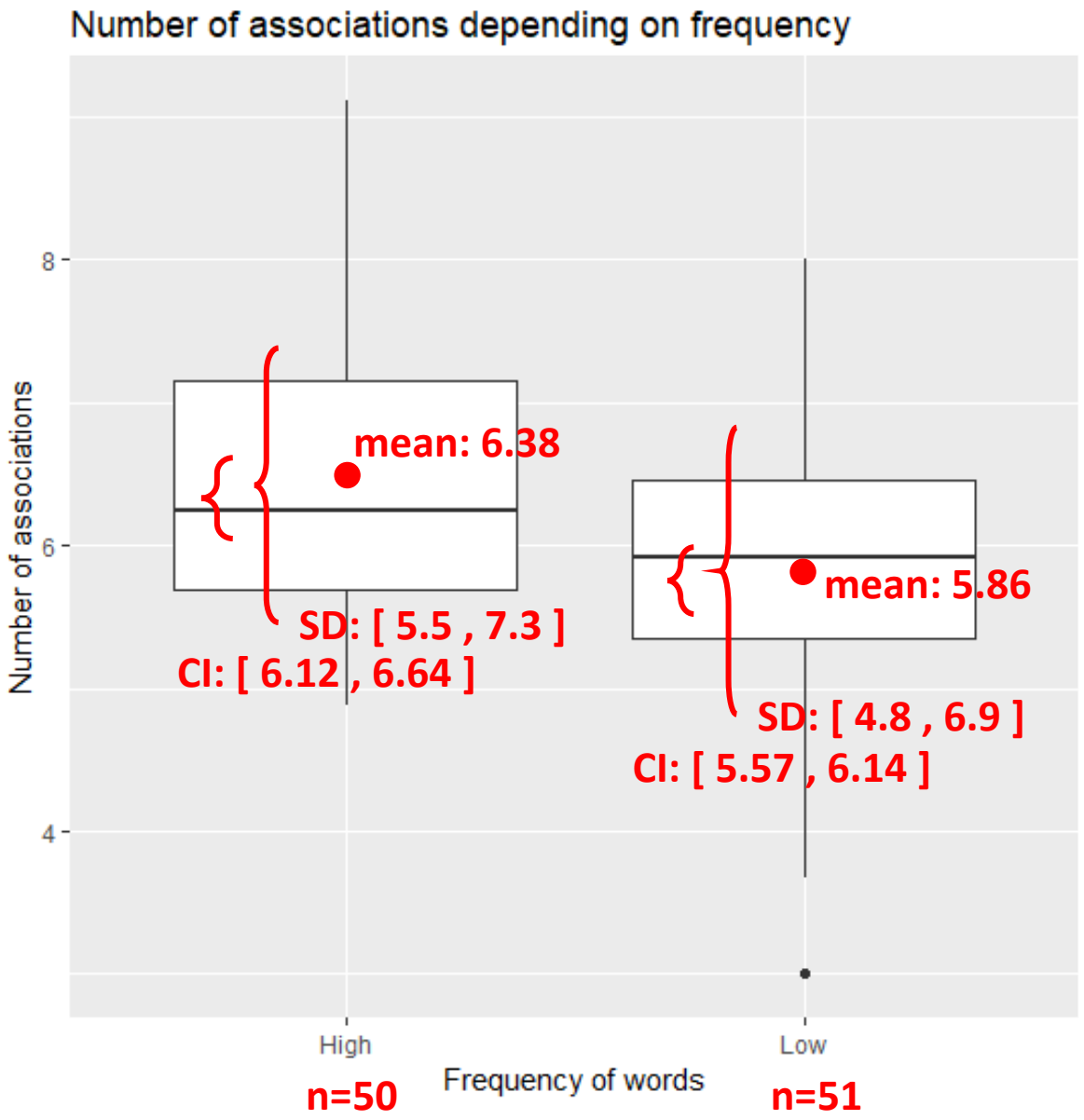
<u>High freq words:</u>

n = 50 words

mean = 6.38
SD = 0.9

mean + SD = 7.3
mean – SD = 5.5

$z_{95\%conf}$ = 1.96
SE = SD/√n = 0.13
CI = mean ± 1.96*SE
CI = 6.4 ± 1.96*0.13
CI = 6.4 ± 0.26
CI = [6.12 , 6.64]



<u>Low freq words:</u>

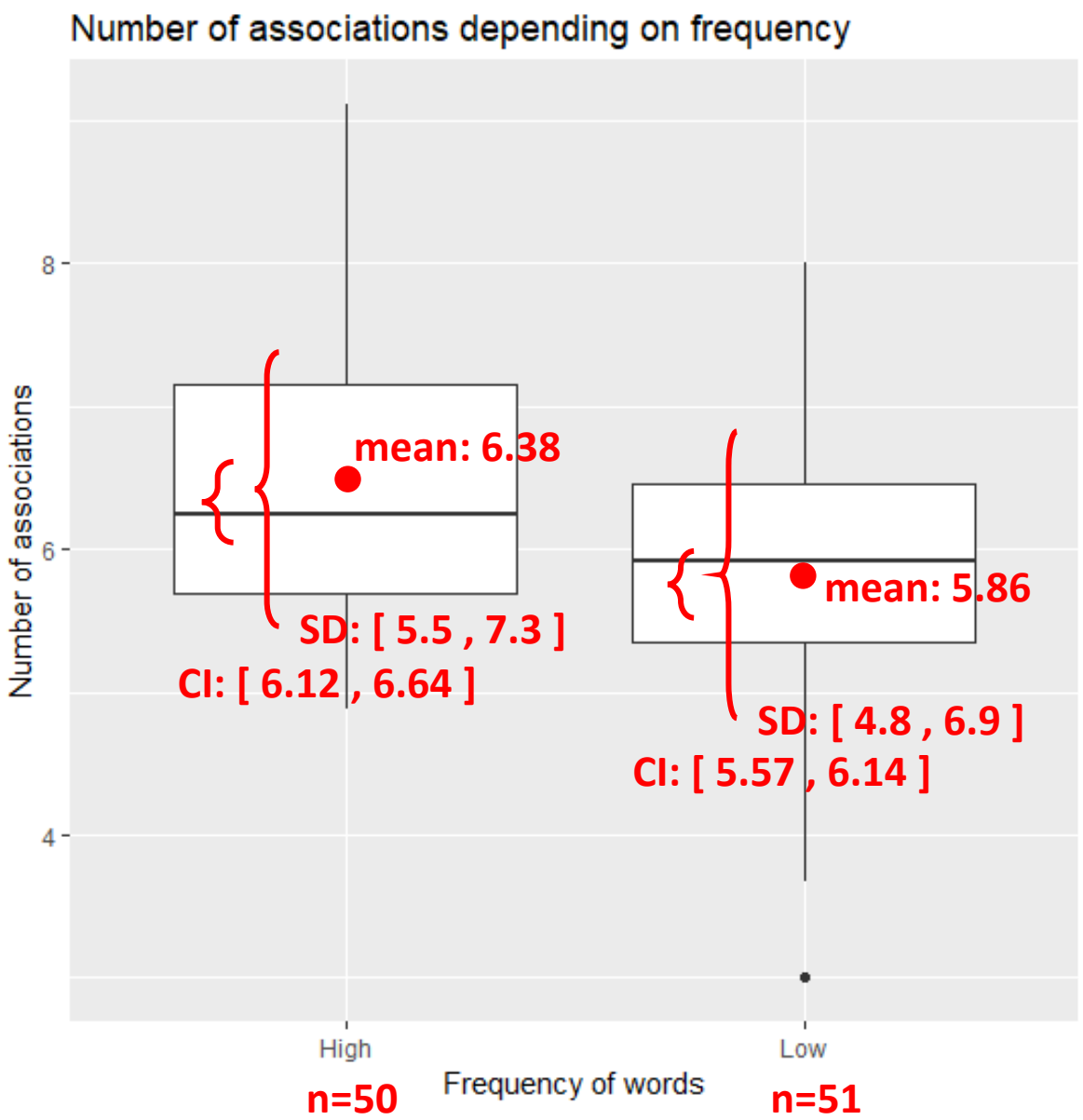
n = 51 words

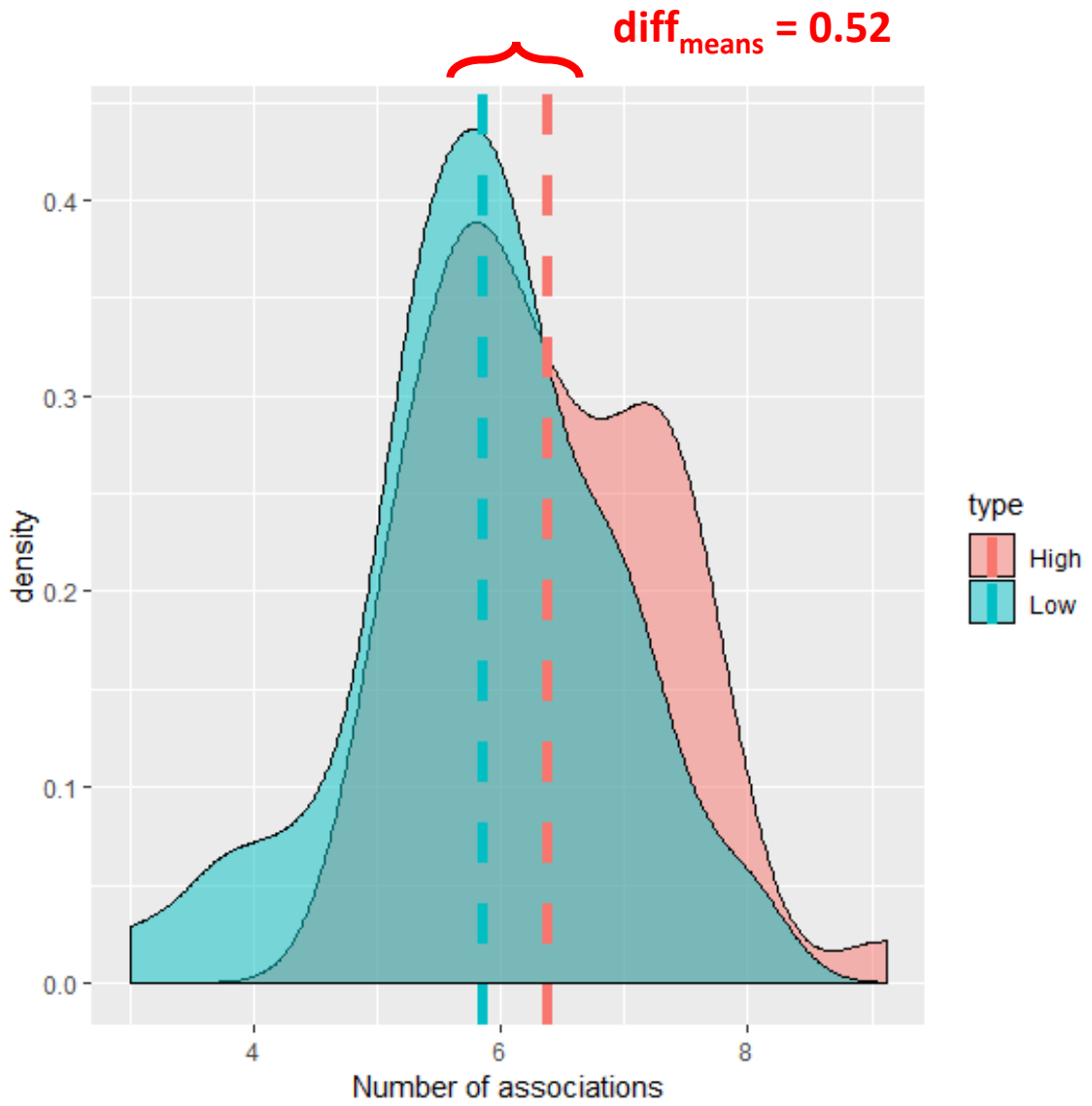
mean = 5.86
SD = 1.03

mean + SD = 6.9
mean – SD = 4.8

$z_{95\%conf}$ = 1.96
SE = SD/√n = 0.14
CI = mean ± 1.96*SE
CI = 5.9 ± 1.96*0.14
CI = 5.9 ± 0.28
CI = [5.57 , 6.14]

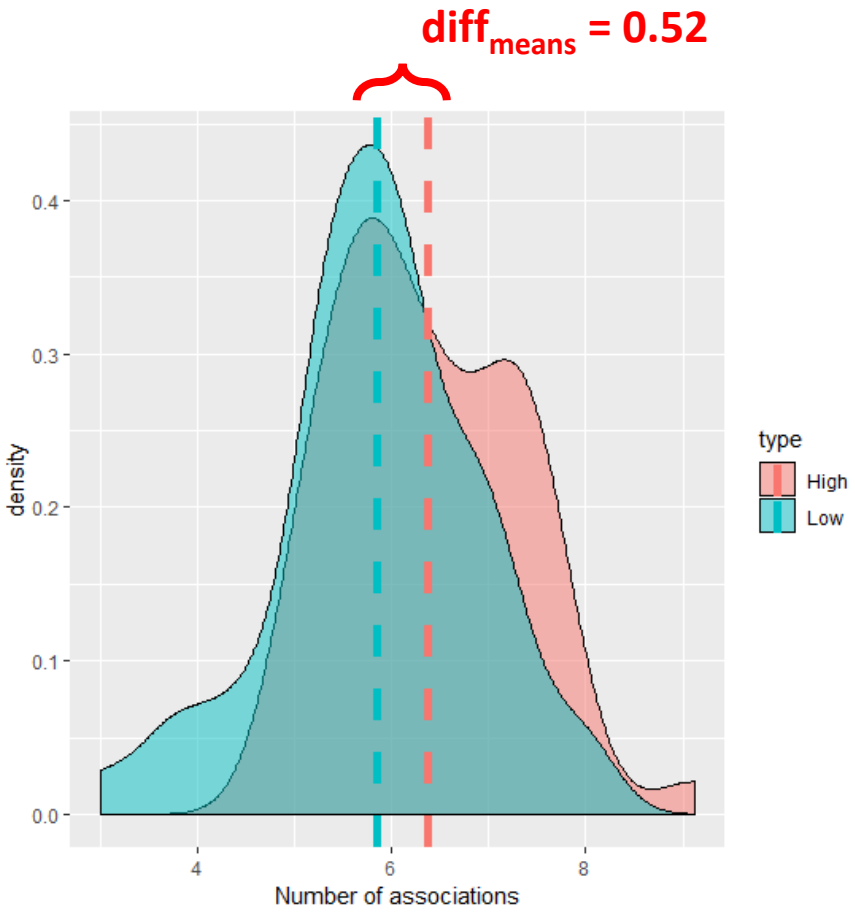
T-test of two independent samples





(73% overlap)

Ho and alternate hypothesis

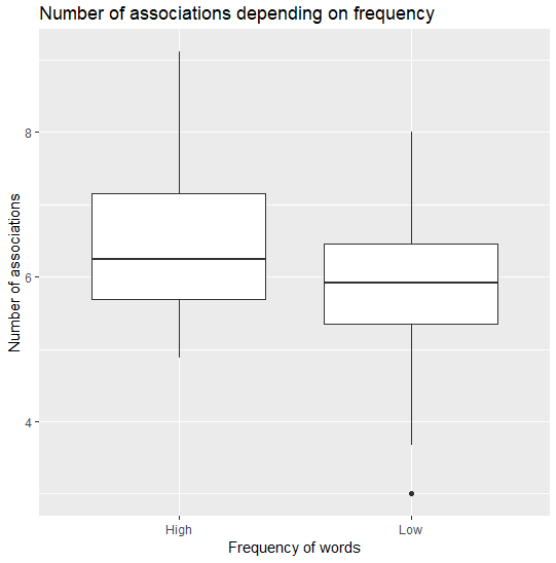


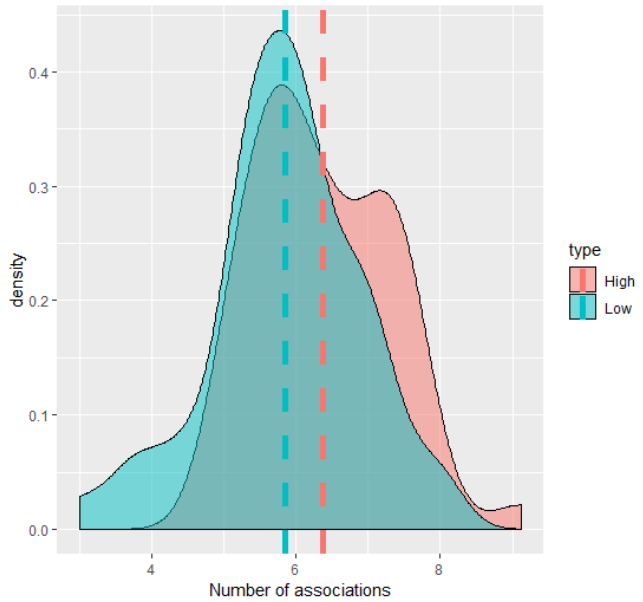
Here, we have two hypotheses:

H_0 (null hypothesis): There is NO real difference in the means of the two groups. It's just as coincidence.

Alternative hypothesis: There really is a difference in the means of the two groups.

T-test of two independent samples



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> t.test(pym_high$assoc, pym_low$assoc)

        Welch Two Sample t-test

data:  pym_high$assoc and pym_low$assoc
t = 2.6717, df = 98.281, p-value = 0.008835

alternative hypothesis: true difference
in means is not equal to 0

95 percent confidence interval:
 0.1344293 0.9106687

sample estimates:
mean of x mean of y
 6.380000  5.857451
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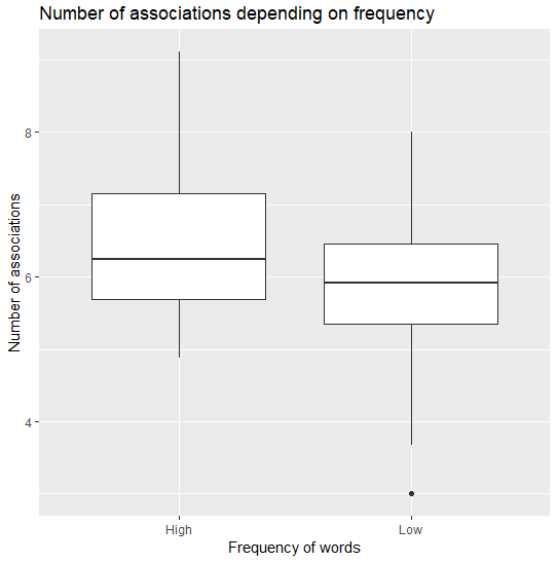

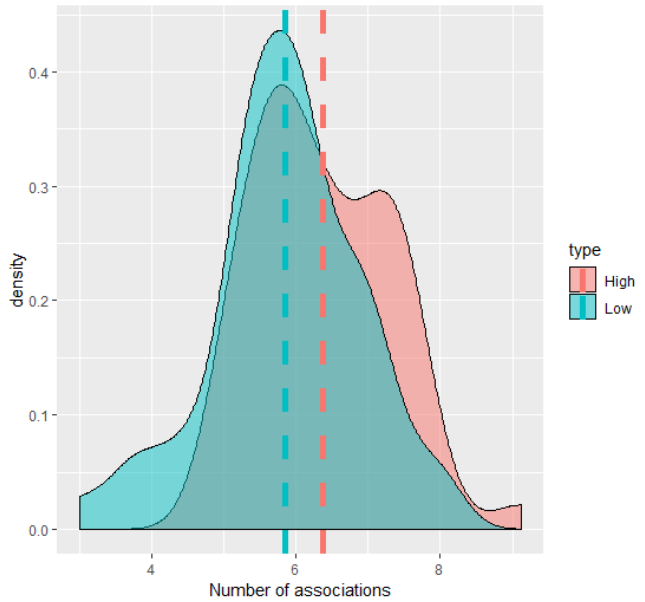


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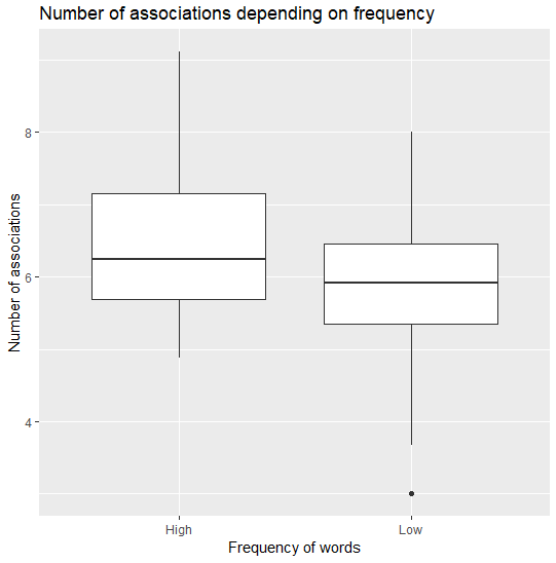
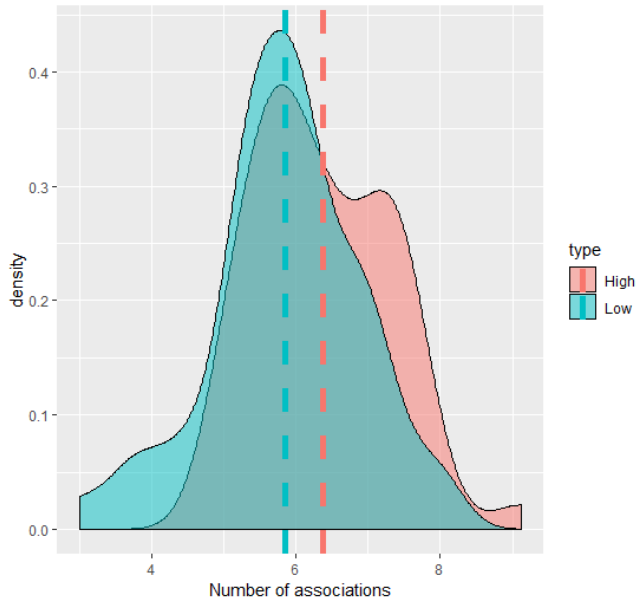
Statement of hypothesis

T-test of two independent samples



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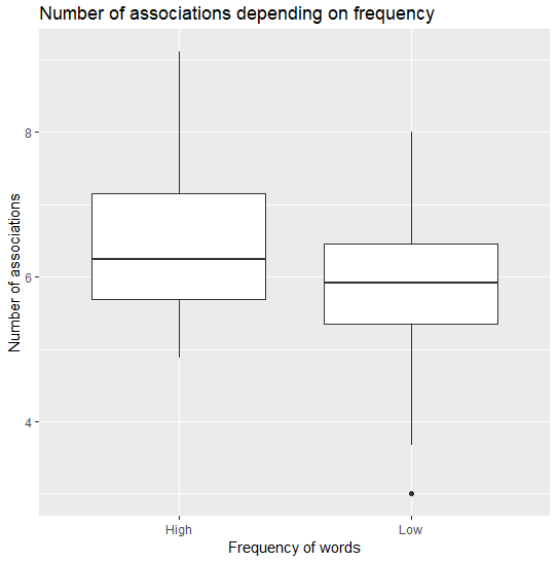
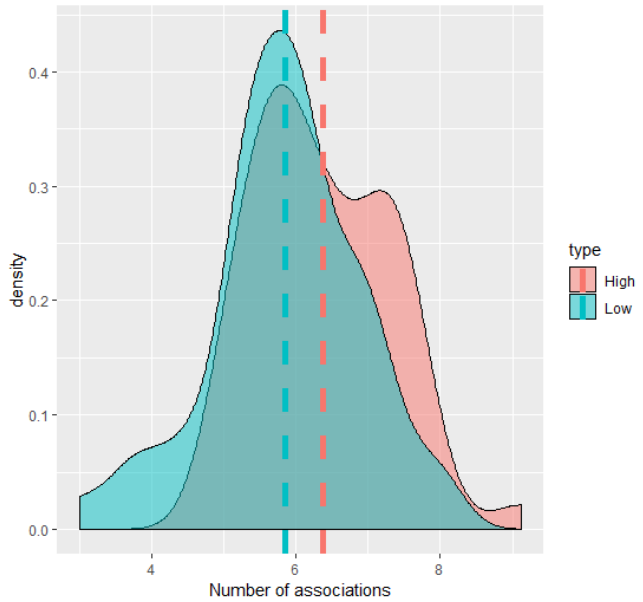
CI for the difference between the means. It does not contain zero!

Means of the first and second group (‘high’ and ‘low’)

T-test of two independent samples



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Degrees of freedom: Approximately the number of total words minus 2

Degrees of Freedom

Let's say you have a set of three numbers:

n_1 n_2 n_3

And you know that their average is 10.

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e.g. 13 2 n_3 (Average = 10)

So, is the third number free to vary?

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So, is the third number free to vary?
No. It needs to be 15 so that our average is 10.
(13+2+15 = 30 30/3 = 10)

You have two **degrees of freedom**.

Degrees of Freedom

Let's say you have a set of three numbers:

n_1 n_2 n_3

And you know that their average is 10.

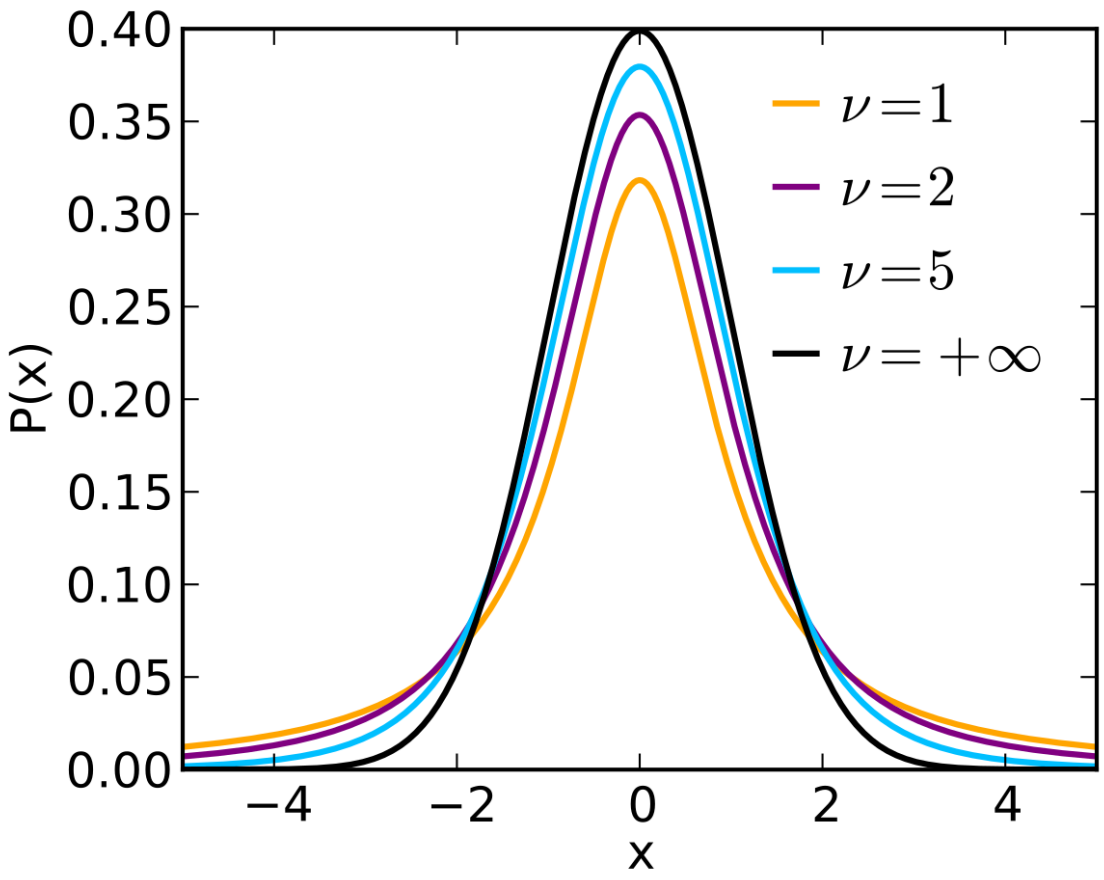
Two of the numbers can very freely:

e.g. 8 13 n_3 (Average = 10)

So, is the third number free to vary?
No. It needs to be 9 so that our average is 10.
(8+13+9 = 30 30/3 = 10)

You have two **degrees of freedom**.

Degrees of Freedom

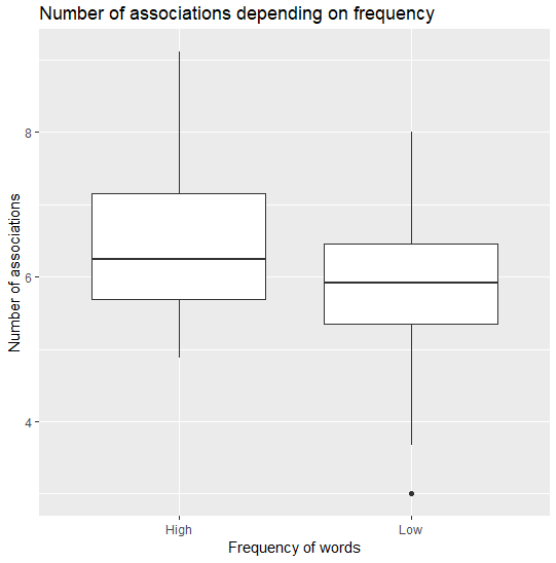


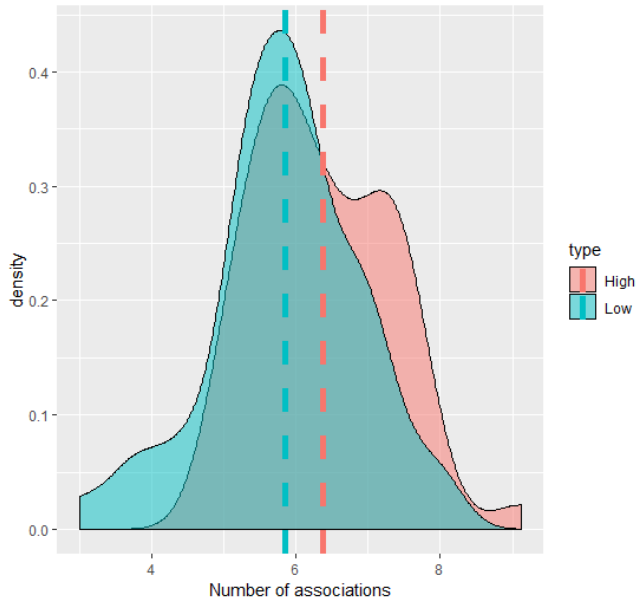
In t-tests, one degree of freedom is spent estimating the mean.

The remaining n-1 degrees of freedom estimate variability.

Therefore, a 1-sample t-test uses a t-distribution with n-1 degrees of freedom

T-test of two independent samples



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Degrees of freedom:
Approximately the number of total words minus 2 (101-2)

If you *really* want to know: It is (n_1+n_2-1-1), but it then undergoes the Welch-Satterthwaite correction for unequal variances.

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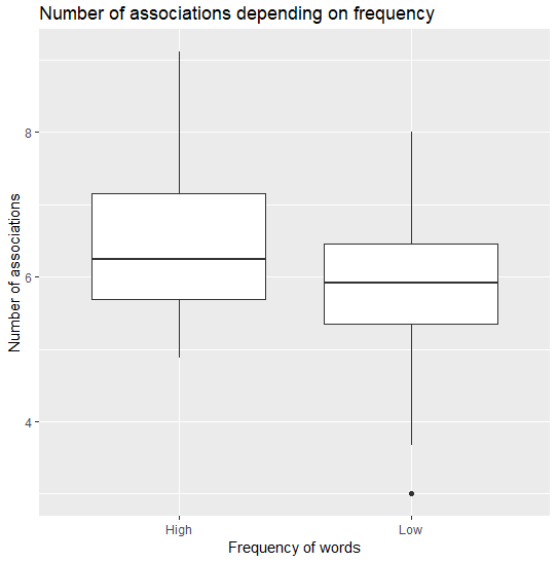

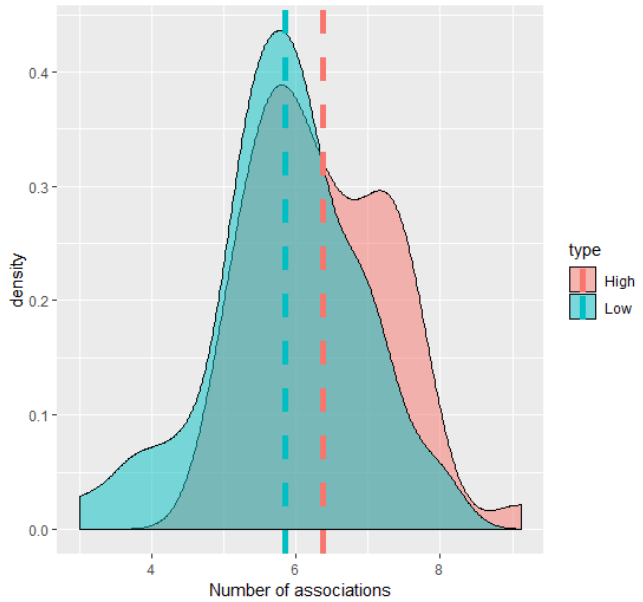


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t:
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This is the difference in means, divided by the standard error of the difference of the means.

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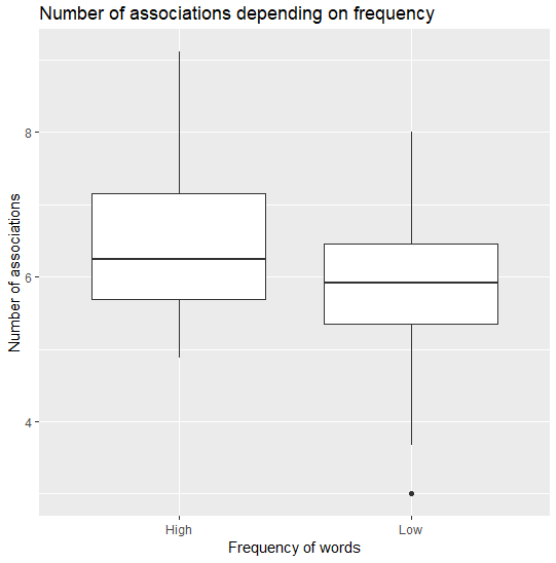

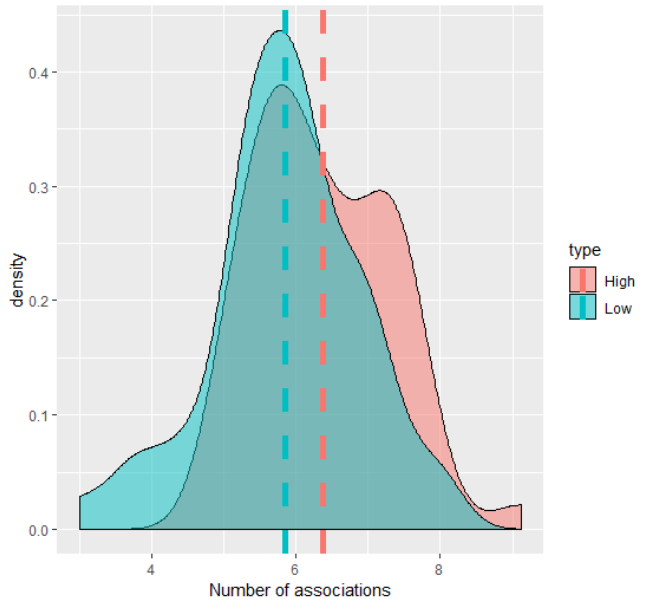


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$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{2}{n}}}$$

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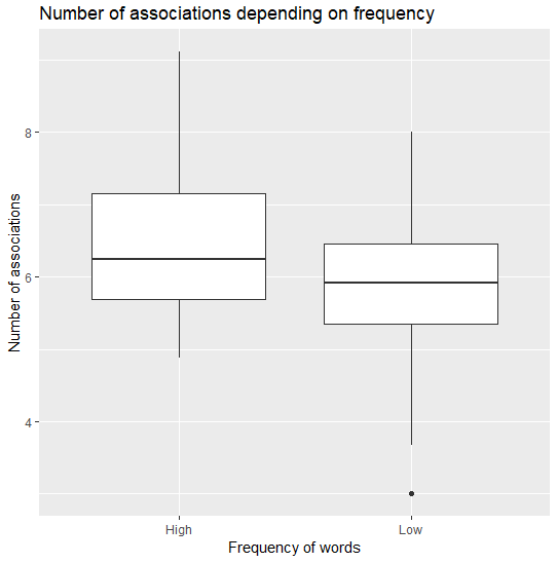

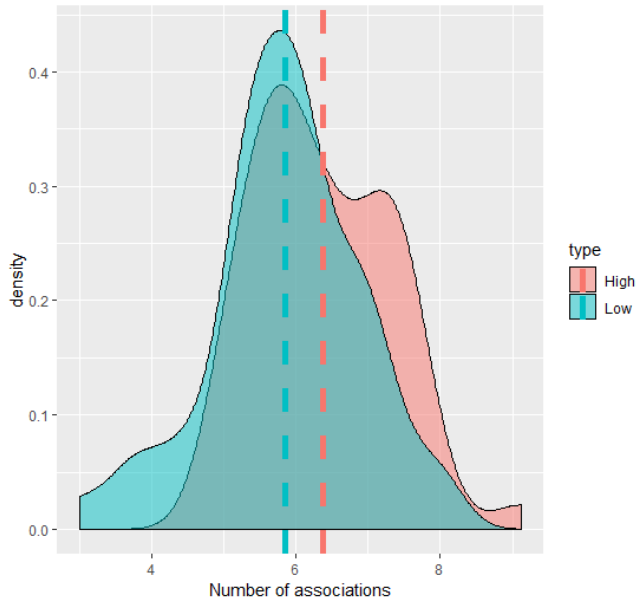


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p-value:
If this value is less than 0.05, then the means are **significantly different**.

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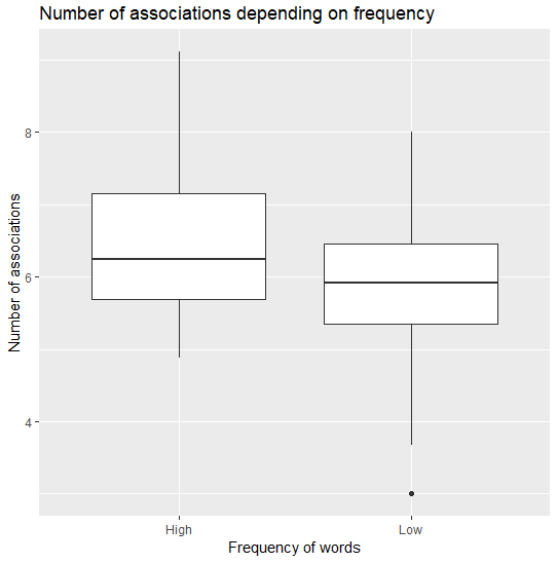

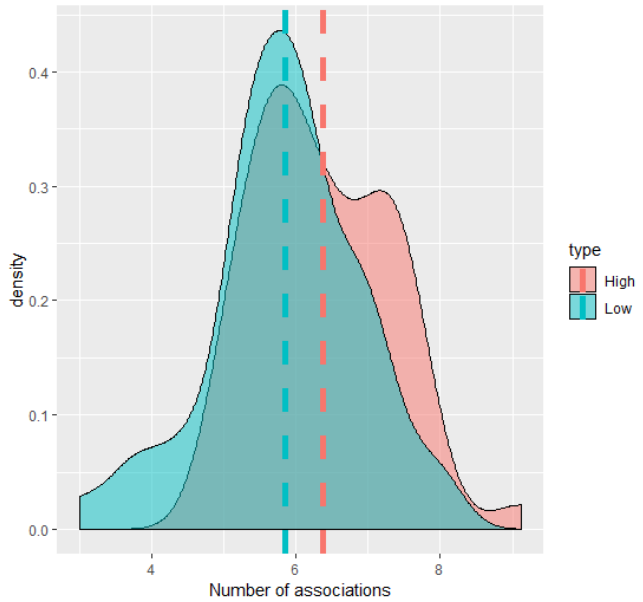


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        Welch Two Sample t-test

data:  pym_high$assoc and pym_low$assoc
t = 2.6717, df = 98.281, p-value = 0.008835

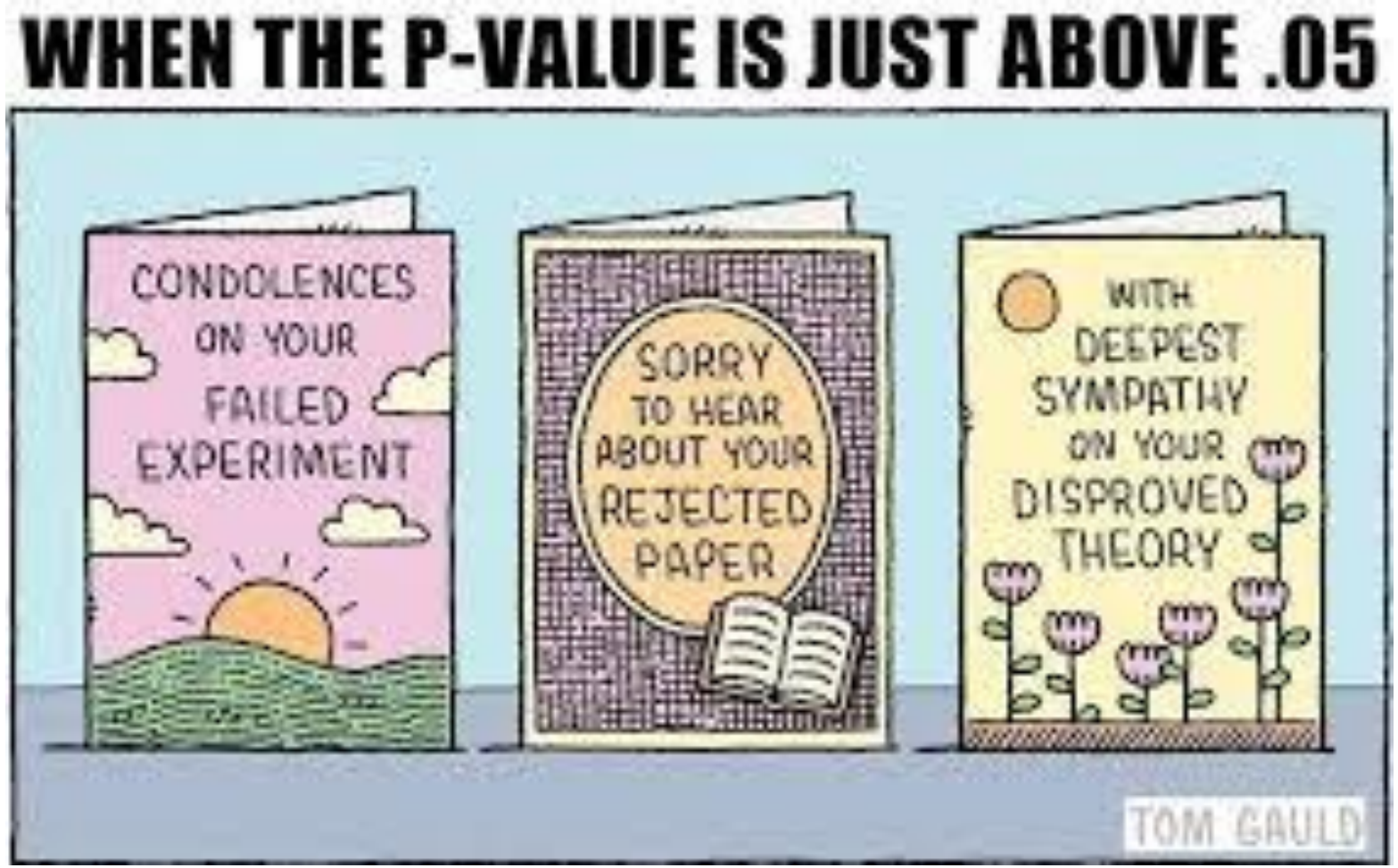
alternative hypothesis: true difference
in means is not equal to 0
```

p-value:

Assuming that the null hypothesis is true (i.e. that the two means are actually the same), the p-value is the probability that this kind of difference (μ_1-μ_2=0.52) will occur at random, if we assume that there is no relationship at all.

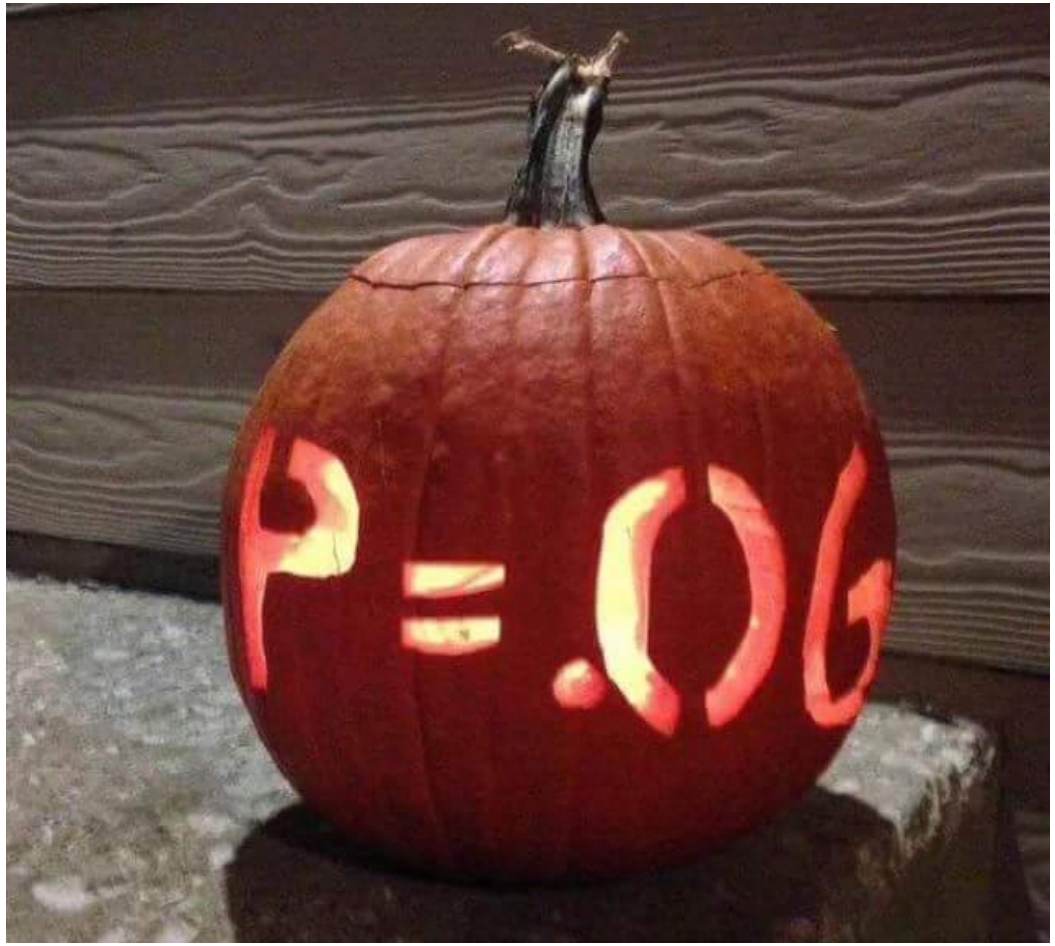
In this case, that probability is 8/1000 = 0.8% (1/125)

P-values



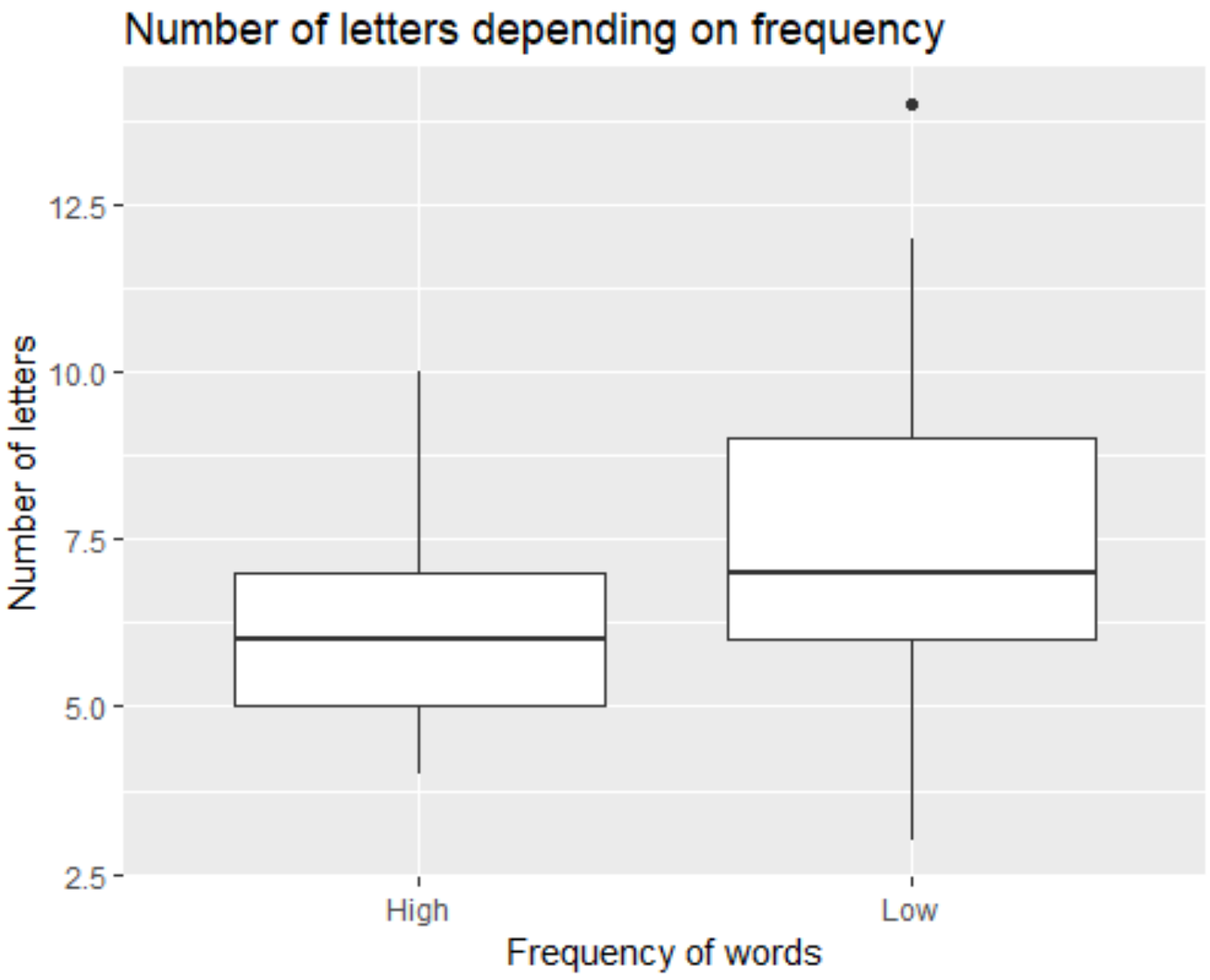
This value is completely arbitrary, btw, and comes from a recommendation from Fischer in the 1920s.

P-values

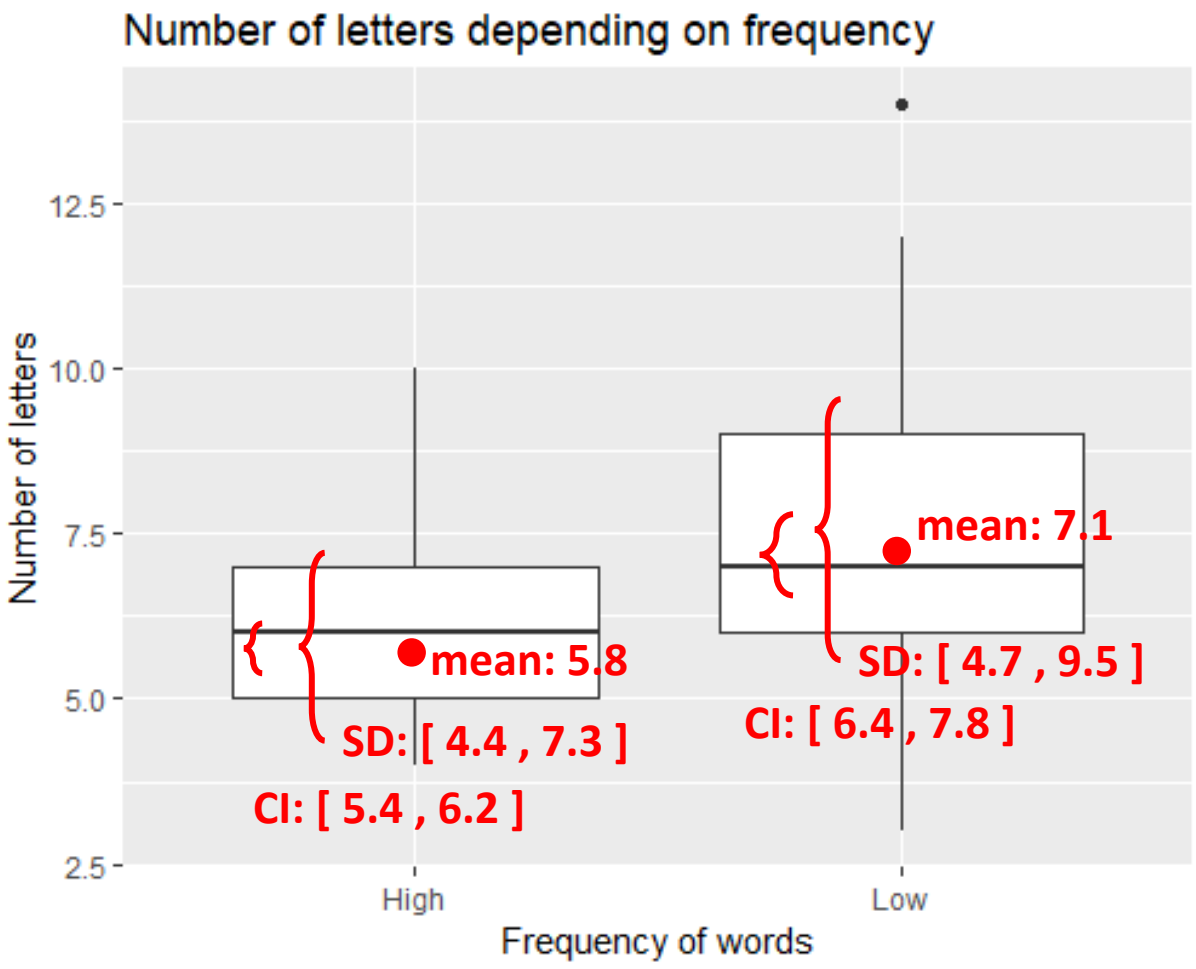


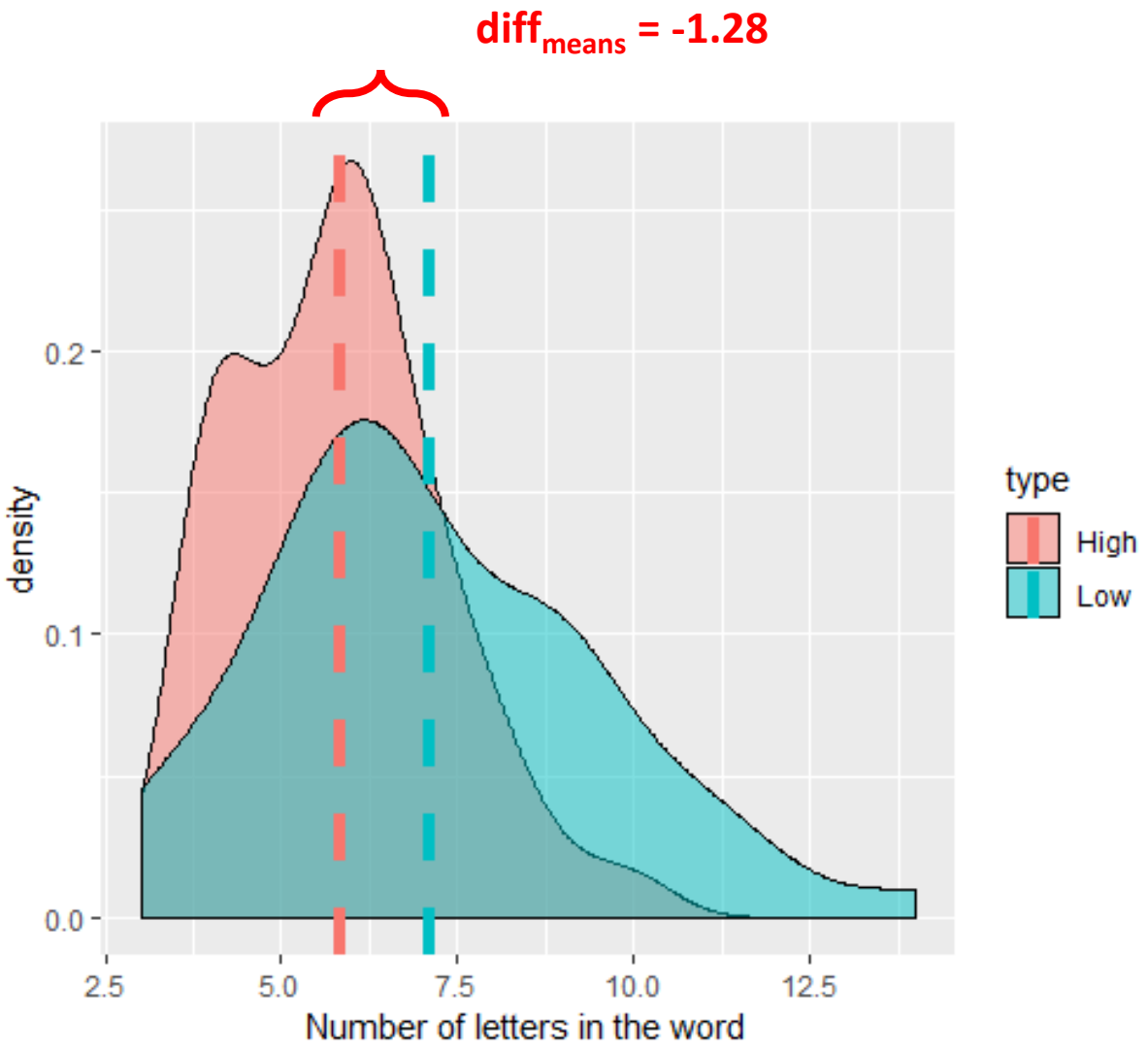
This value is completely arbitrary, btw, and comes from a recommendation from Fischer in the 1920s.

T-test of two independent samples



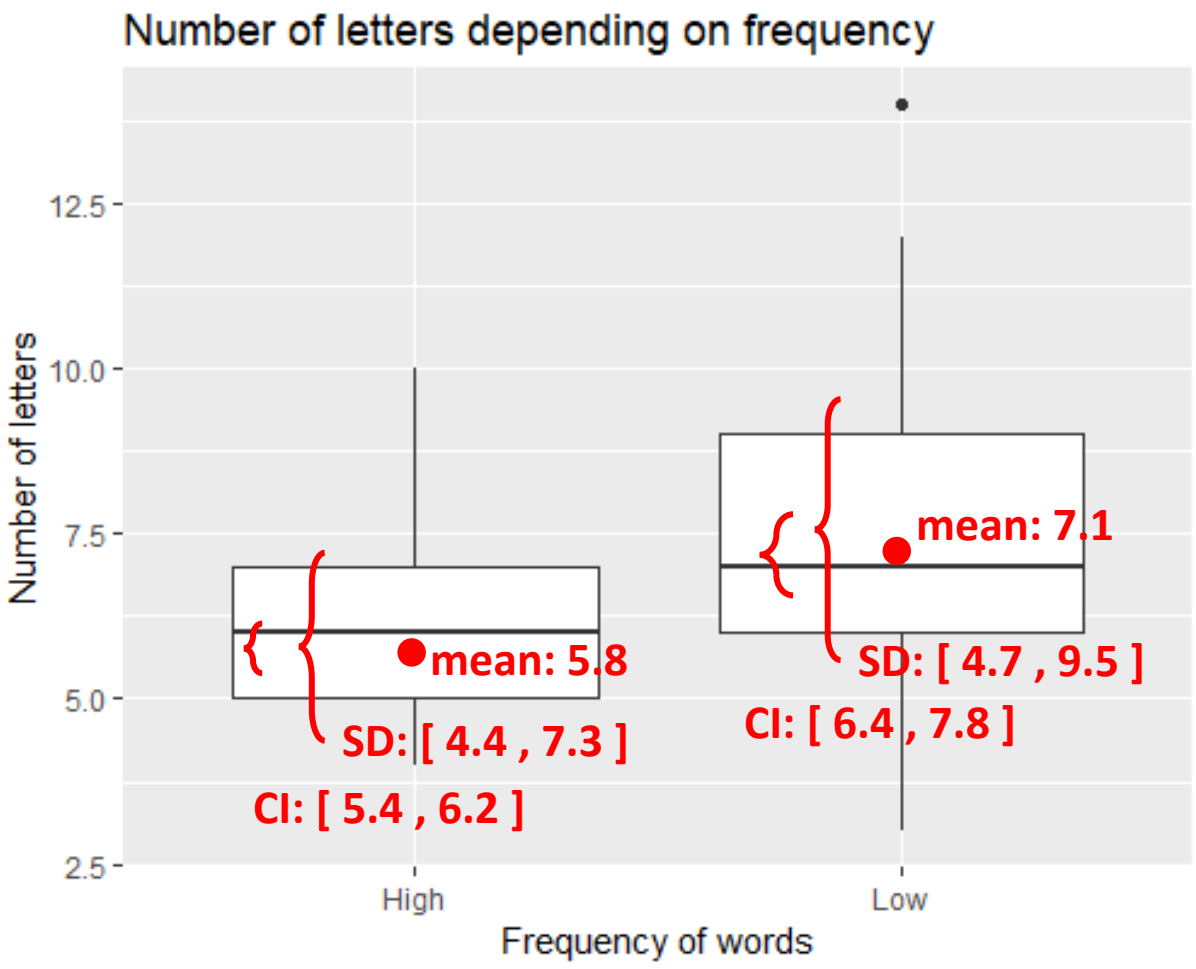
T-test of two independent samples

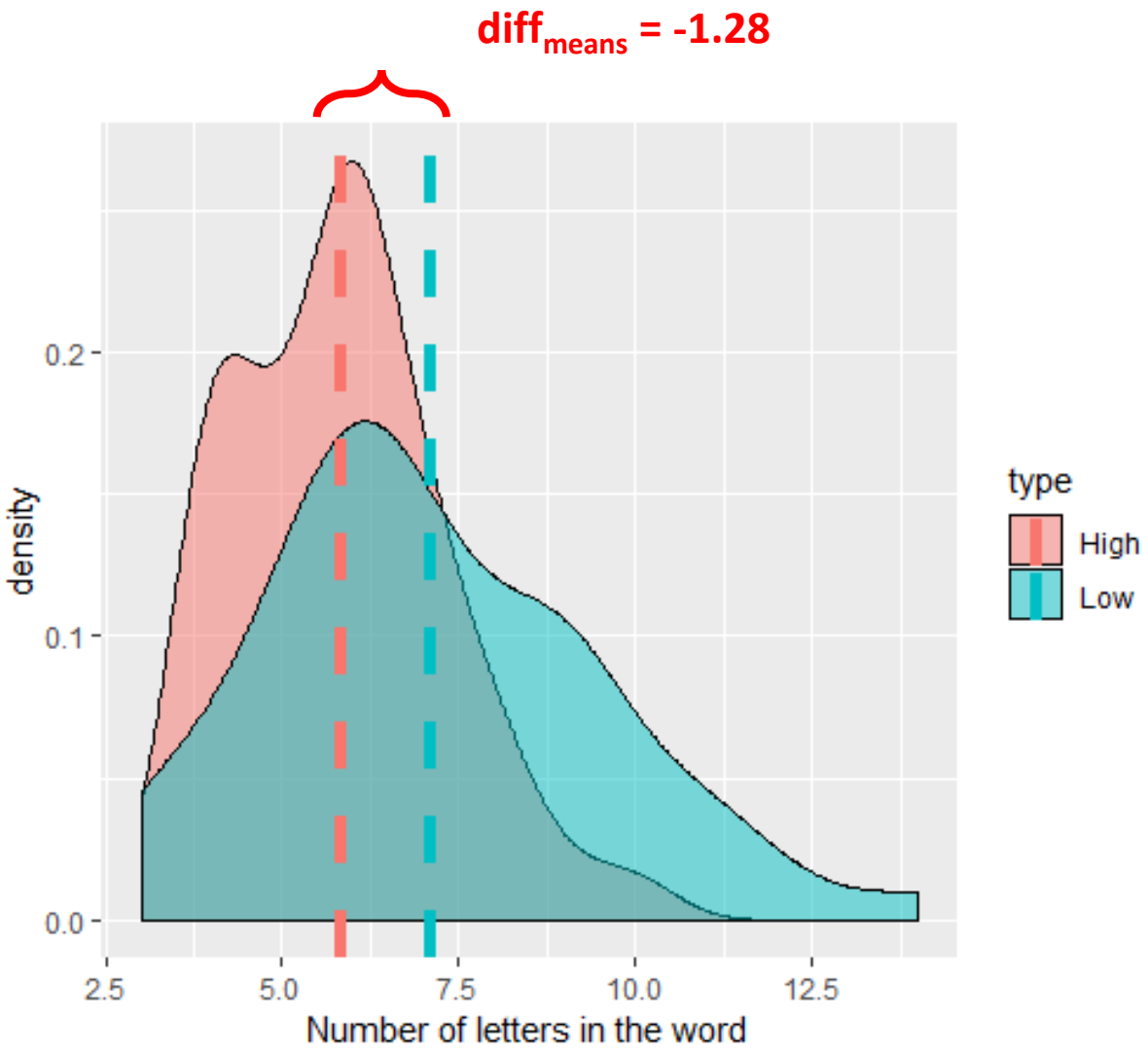




(56% overlap)

T-test of two independent samples

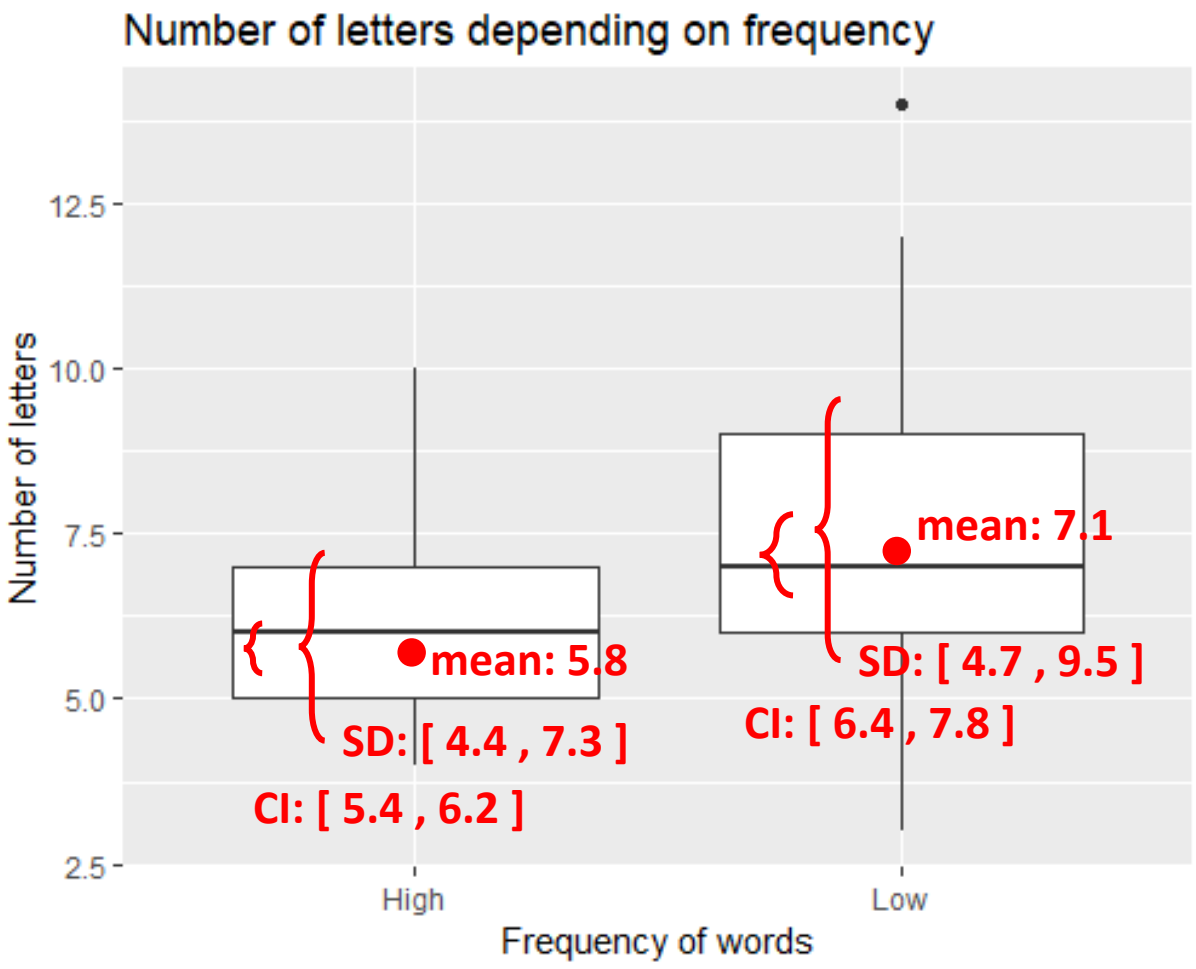


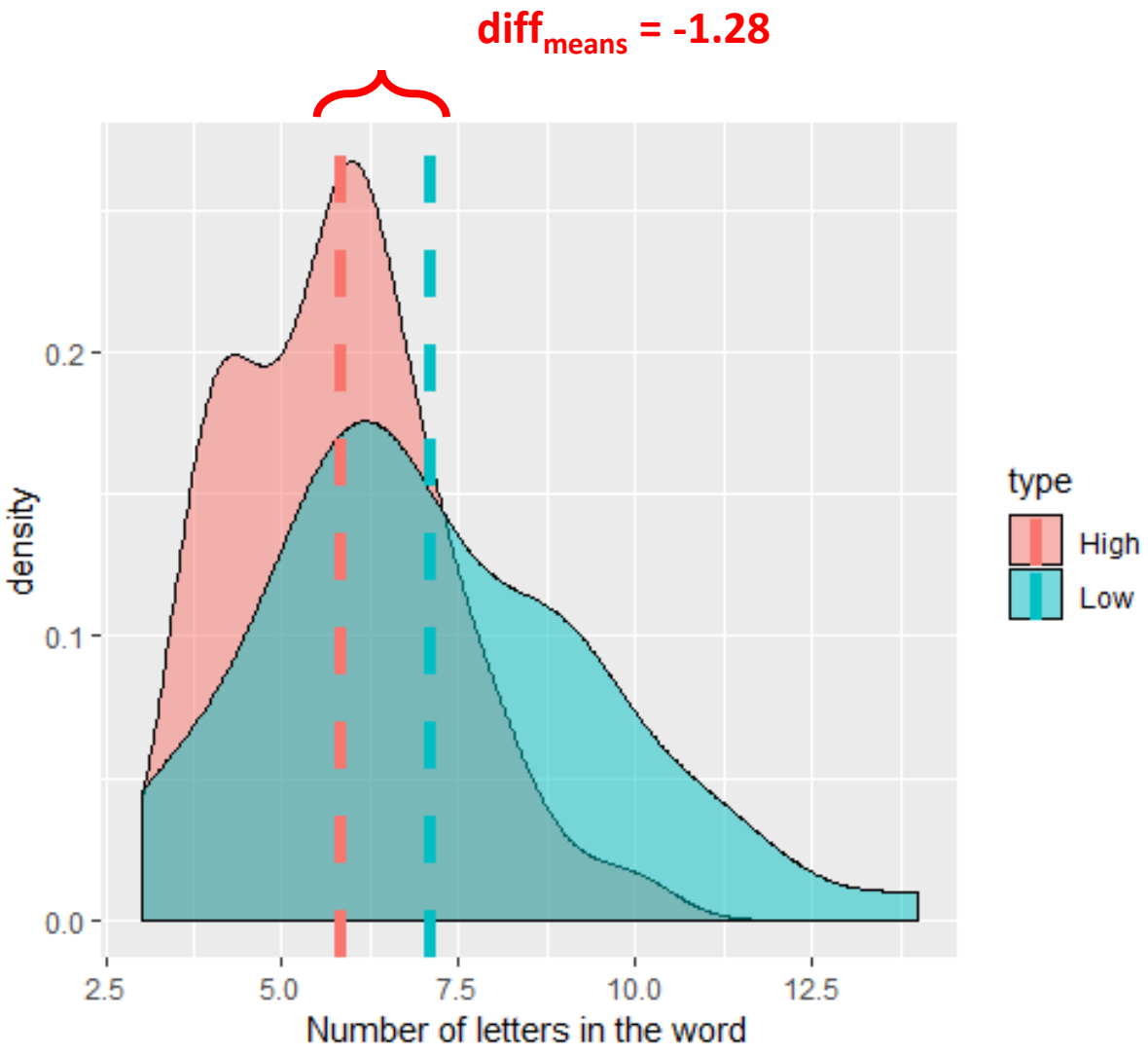


Do the confidence intervals overlap?

(56% overlap)

T-test of two independent samples

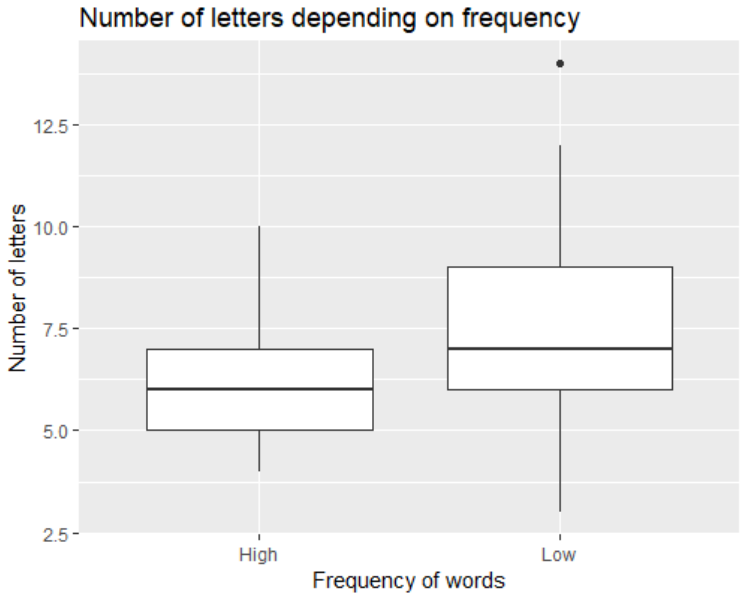


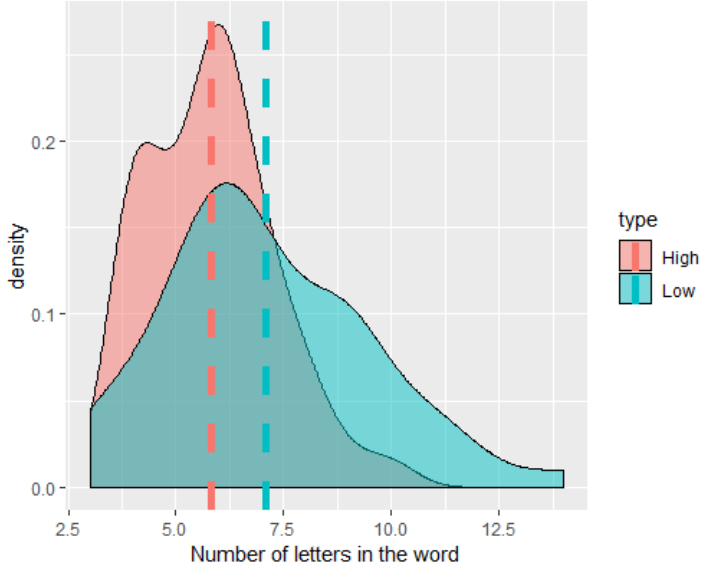


CIs do not overlap. Therefore, we can be fairly sure that the means are different.

(56% overlap)

T-test of two independent samples






```
> t.test(pym_high$let,pym_low$let)

        Welch Two Sample t-test

data:  pym_high$let and pym_low$let
t = -3.2509, df = 82.705, p-value = 0.001665

alternative hypothesis: true difference
in means is not equal to 0

95 percent confidence interval:
 -2.0600041 -0.4960743

sample estimates:
mean of x mean of y
 5.820000  7.098039
```

Tails

Comparing whether two means are equal:

```
t.test(pym_high$let,pym_low$let)
t.test(pym_high$let,pym_low$let, alternative = "two.sided")
```

Comparing whether the first mean is greater than the second:

```
t.test(pym_high$let,pym_low$let, alternative = "greater")
```

Comparing whether the first mean is less than the second:

```
t.test(pym_high$let,pym_low$let, alternative = "less")
```

Tails

```
t.test(pym_high$let,pym_low$let, alternative = "two.sided")

     Welch Two Sample t-test

data:  pym_high$let and pym_low$let
t = -3.2509, df = 82.705, p-value = 0.001665
```

```
t.test(pym_high$let,pym_low$let, alternative = "less")

     Welch Two Sample t-test

data:  pym_high$let and pym_low$let
t = -3.2509, df = 82.705, p-value = 0.0008325
```

Tails

```
t.test(pym_high$let,pym_low$let, alternative = "two.sided")

     Welch Two Sample t-test

data:  pym_high$let and pym_low$let
t = -3.2509, df = 82.705, p-value = 0.001665
```

```
t.test(pym_high$let,pym_low$let, alternative = "less")

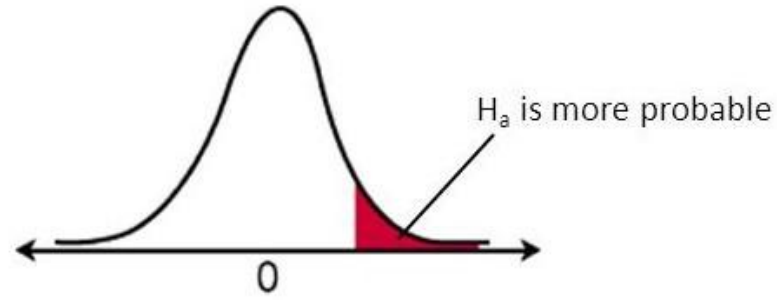
     Welch Two Sample t-test

data:  pym_high$let and pym_low$let
t = -3.2509, df = 82.705, p-value = 0.0008325
```

The p-value in the one-tailed test is half of the p-value in the two-tailed test.

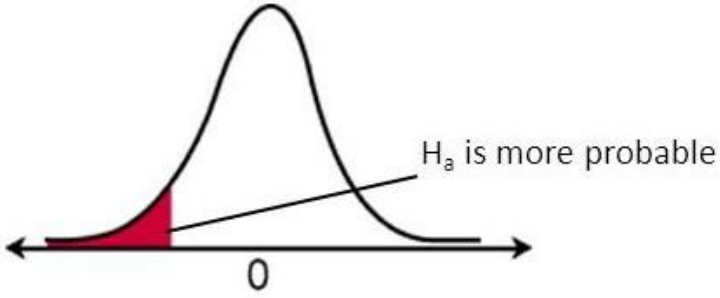
Tails

Your t-tests can have three configurations:



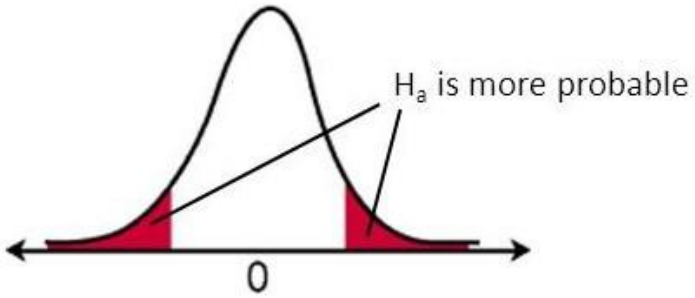
Right-tail test

H_a: $\mu >$ value



Left-tail test

H_a: $\mu <$ value



Two-tail test

H_a: $\mu \neq$ value

Assumptions

There are several assumptioms about how your data needs to be:

- It needs to be **randomly selected**.
- Both groups needs to be **independent** from one another.
- The variables must be **numerical** (at least ordinal scale)
- The data must be **normally distributed**, or you must have at least 30 samples for each group.

Types of Errors

		Reality	
		True	False
Measured or Perceived	True	Correct ☺	Type 1 error False Positive
	False	Type 2 error False Negative	Correct ☺

Power

The p-value is the probability that there is a difference in the sample means, **assuming the null** hypothesis is true.

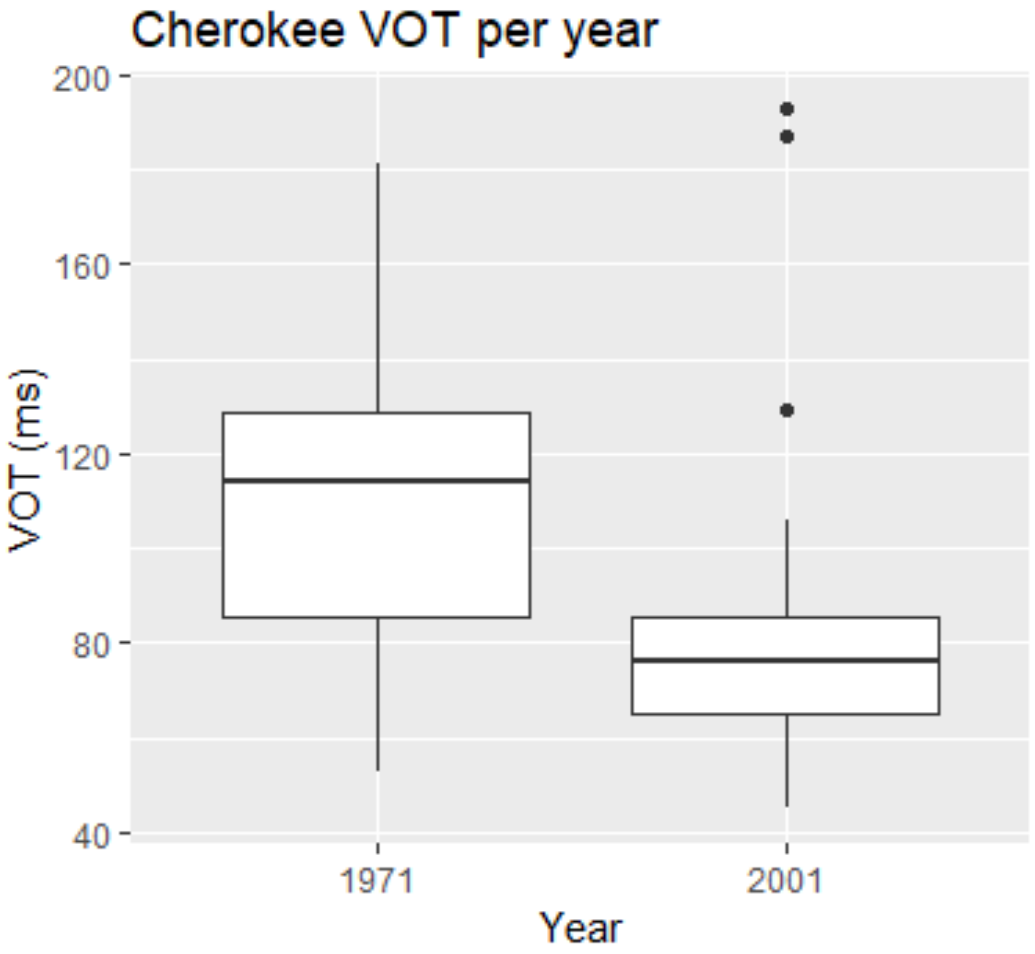
The **power** is the probability of finding a difference in the sample means, **assuming the alternative** hypothesis is true. (in other words, assuming there *is* an effect to find)

Types of Errors

Measured or Perceived	Reality: True	Reality: False
True	**Correct** (true positive) Power = 1-β = 80%	**Type 1 error** False Positive
False	**Type 2 error** False Negative Not enough **power** β = 20%	**Correct** (true negative)

1-β = power = 0.80

Power



1971: n=16. Shapiro Test: p=0.75

2001: n=26. Let's asume it's normal.

```
t.test(vot1971$VOT, vot2001$VOT)

        Welch Two Sample t-test

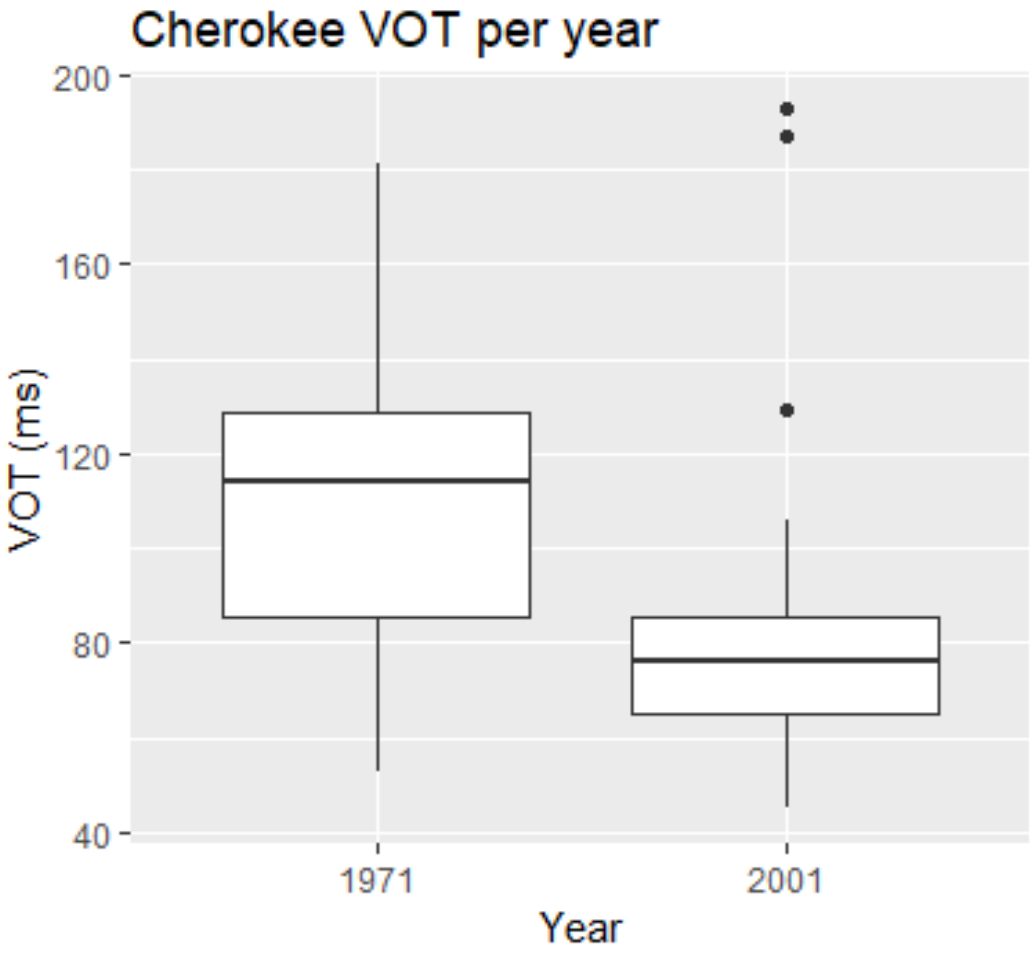
data:  vot1971$VOT and vot2001$VOT
t = 2.6137, df = 36.825, p-value = 0.0129

alternative hypothesis: true difference in
means is not equal to 0

95 percent confidence interval:
  6.480602 51.211706

sample estimates:
mean of x mean of y
113.50000  84.65385
```

Power




```
power.t.test(power=0.80, sig.level=0.05,
delta=29, sd=36, type="two.sample",
alternative="two.sided")
```

```
     Two-sample t test power calculation

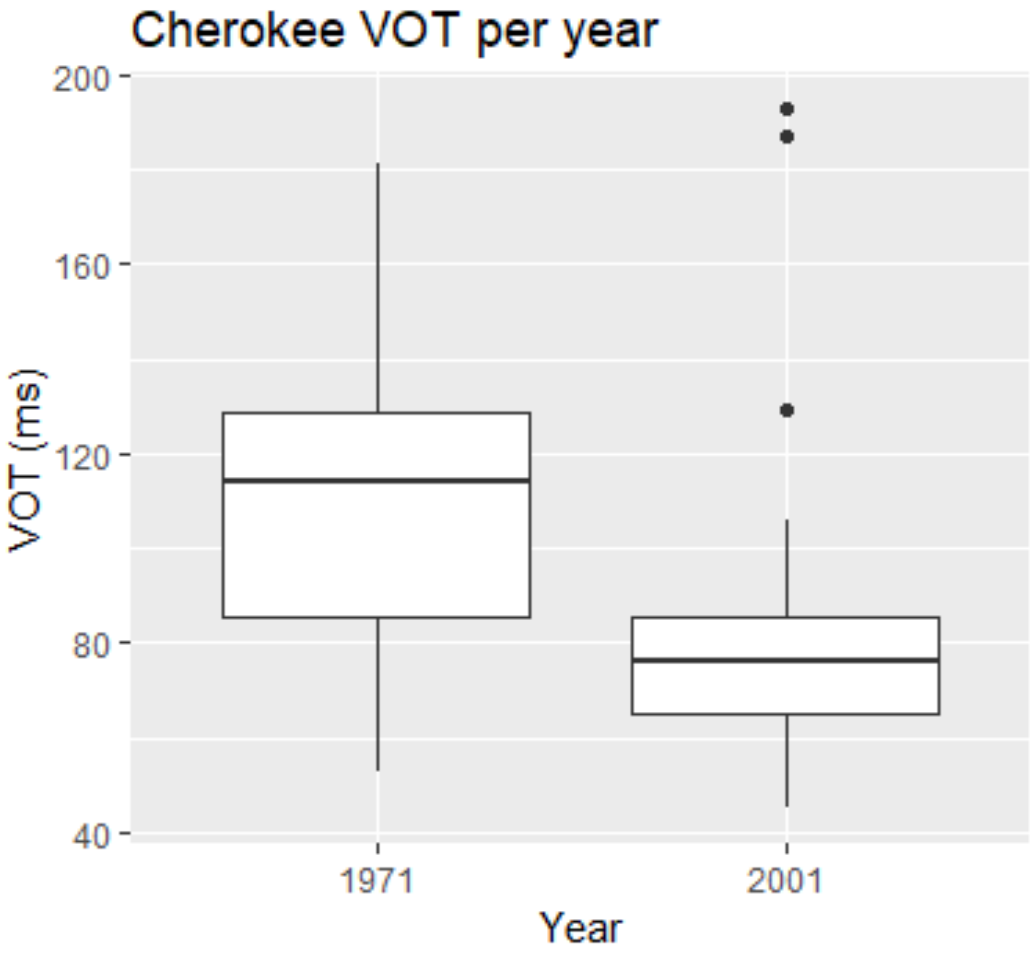
              n = 25.18799
          delta = 29
             sd = 36
      sig.level = 0.05
          power = 0.8
    alternative = two.sided
```

1971: n=16. Shapiro Test: p=0.75

2001: n=26. Let's asume it's normal.

NOTE: n is number in *each* group

Power




```
power.t.test(power=0.60, sig.level=0.05,
delta=29, sd=36, type="two.sample",
alternative="two.sided")
```

```
     Two-sample t test power calculation

              n = 16.11108
          delta = 29
             sd = 36
      sig.level = 0.05
          power = 0.6
    alternative = two.sided
```

1971: n=16. Shapiro Test: p=0.75

2001: n=26. Let's asume it's normal.

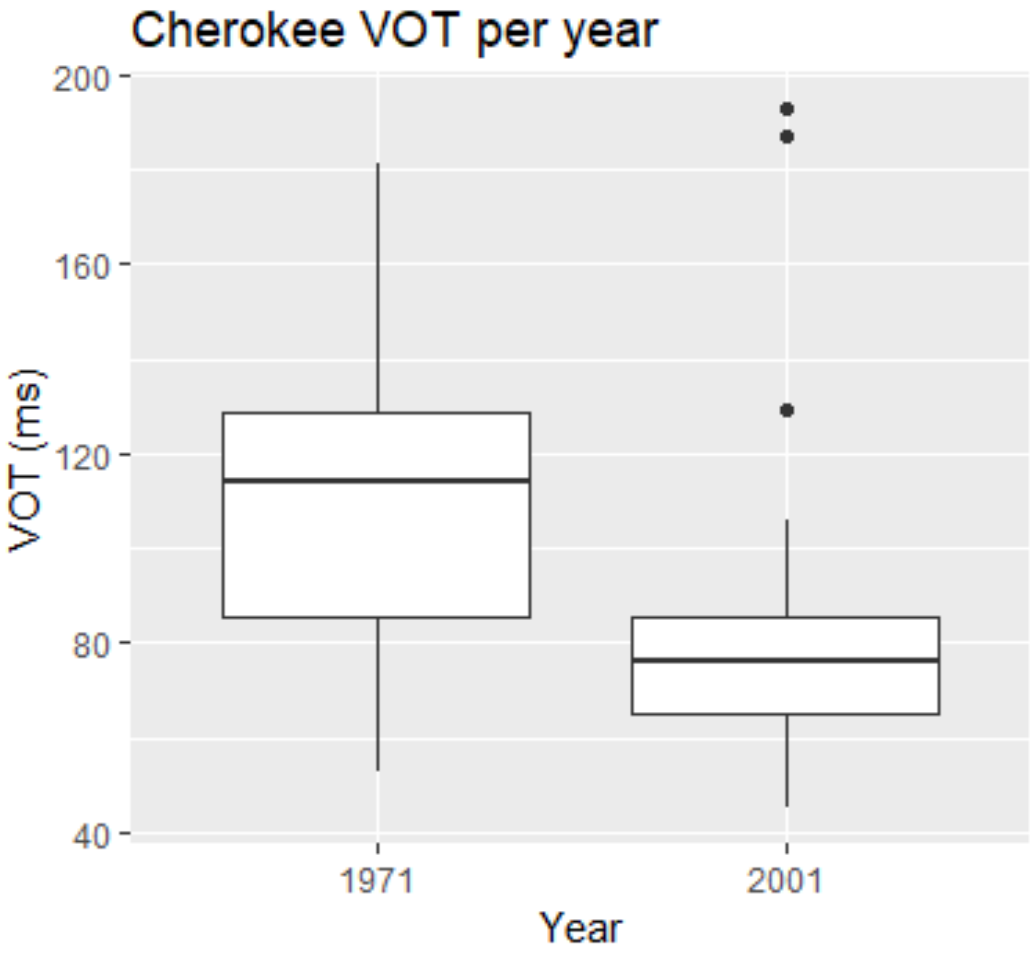
NOTE: n is number in *each* group

Types of Errors

		Reality	
		True	False
Measured or Perceived	True	**Correct** (true positive) Power = 1-β = 60%	**Type 1 error** False Positive
	False	**Type 2 error** False Negative Not enough **power** β = 40%	**Correct** (true negative)

1-β = power = 0.60

Power




```
power.t.test(power=0.95, sig.level=0.05,
delta=29, sd=36, type="two.sample",
alternative="two.sided")
```

```
     Two-sample t test power calculation

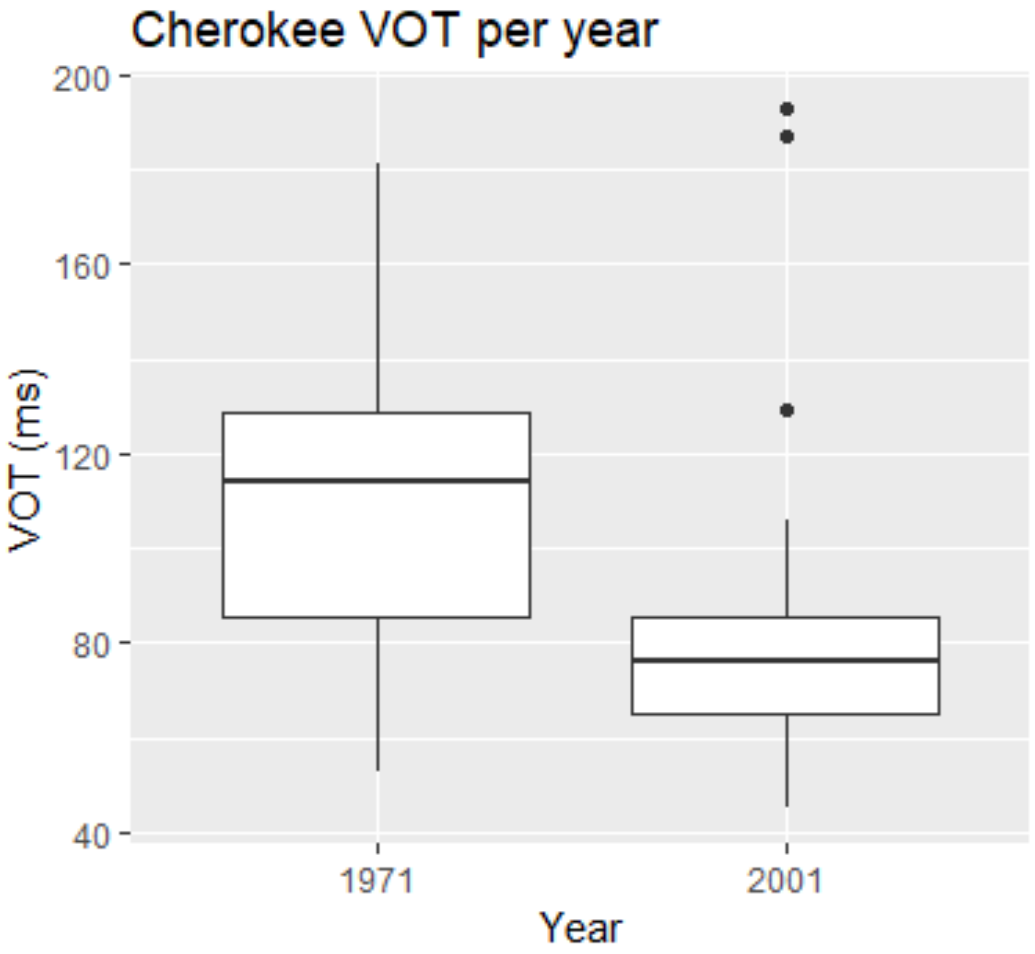
              n = 41.0363
          delta = 29
             sd = 36
      sig.level = 0.05
          power = 0.95
    alternative = two.sided
```

1971: n=16. Shapiro Test: p=0.75

2001: n=26. Let's asume it's normal.

NOTE: n is number in *each* group

Power




```
power.t.test(power=0.99, sig.level=0.05,
delta=29, sd=36, type="two.sample",
alternative="two.sided")
```

```
     Two-sample t test power calculation

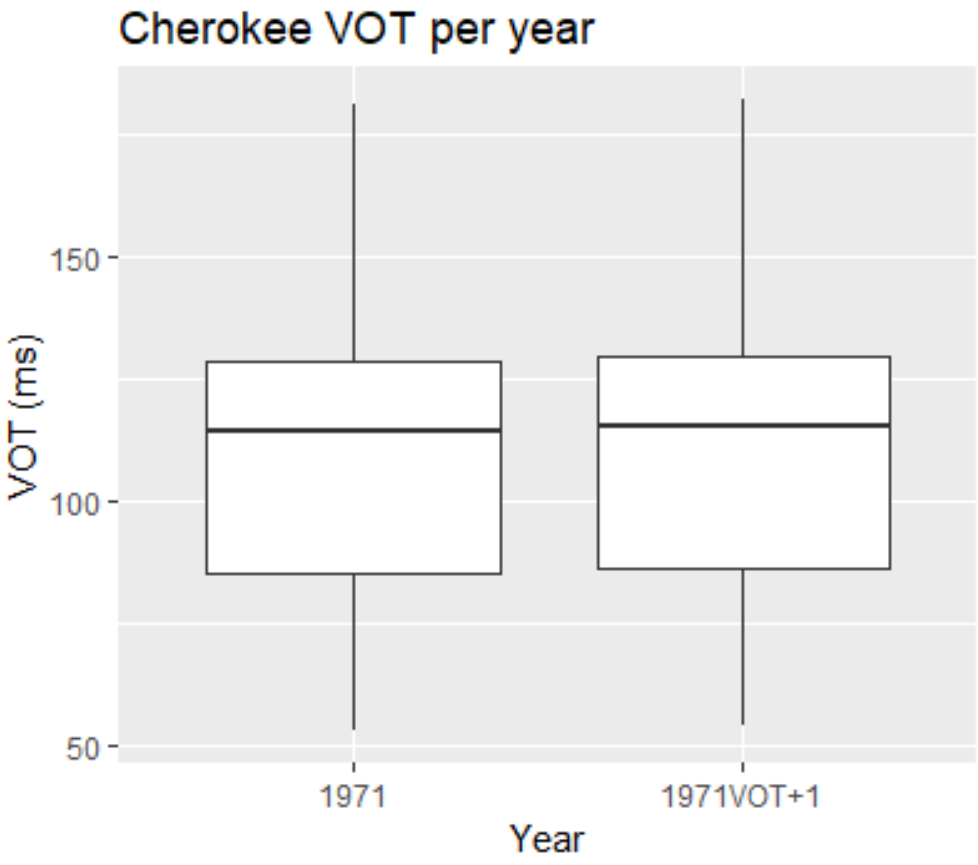
              n = 57.60523
          delta = 29
             sd = 36
      sig.level = 0.05
          power = 0.99
    alternative = two.sided
```

1971: n=16. Shapiro Test: p=0.75

2001: n=26. Let's asume it's normal.

NOTE: n is number in *each* group

Power




```
power.t.test(power=0.8, sig.level=0.05,
delta=1, sd=36, type="two.sample",
alternative="two.sided")
```

```
     Two-sample t test power calculation

              n = 20345.26
          delta = 1
             sd = 36
      sig.level = 0.05
          power = 0.8
    alternative = two.sided
```

1971: n=16. Shapiro Test: p=0.75

1971': n=16. (Same, but +1 of 1971)

```
NOTE: n is number in *each* group
```

Types of Errors

The **power** is the probability of finding a difference in the sample means, **assuming the alternative** hypothesis is true. (in other words, assuming there *is* an effect to find)

		Reality	
		True	False
Measured or Perceived	True	Correct (true positive) Power = 1-β = 80%	Type 1 error False Positive
	False	Type 2 error False Negative Not enough power β = 20%	Correct (true negative)

Paired Samples

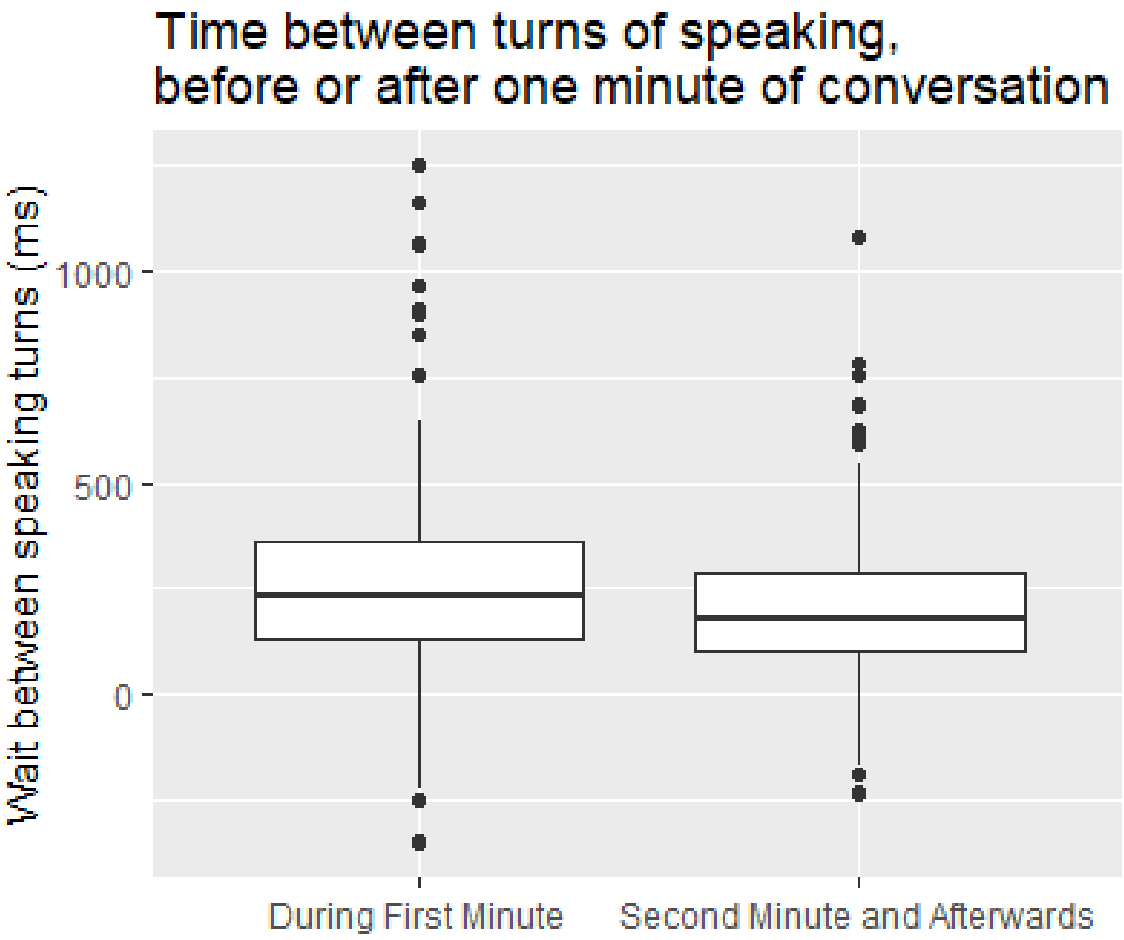
So far we have looked at **unpaired samples** (where the tokens in the two groups are not related).

In **paired samples**, the two samples are connected:

- The test-score on a proficiency test before AND after two semesters
- The VOT of a person before AND after six months

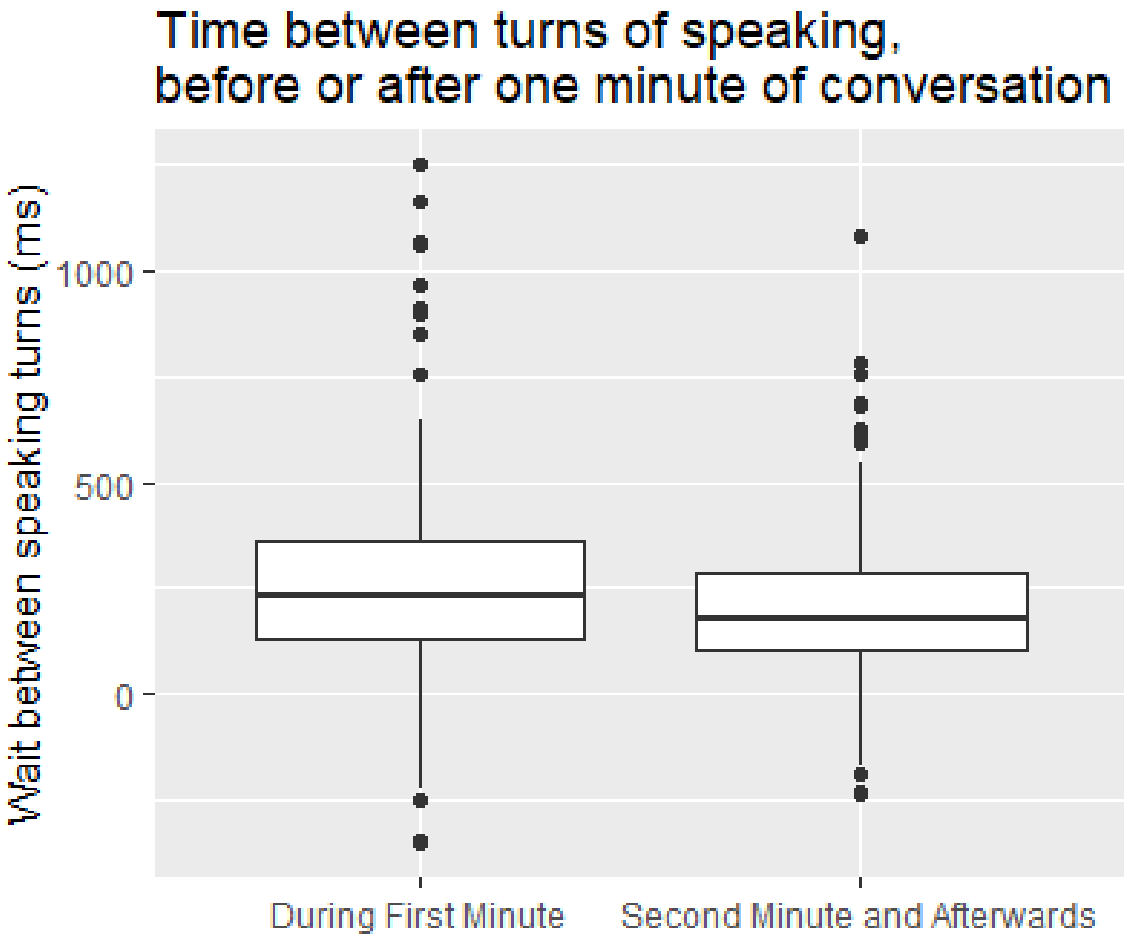
Paired Samples

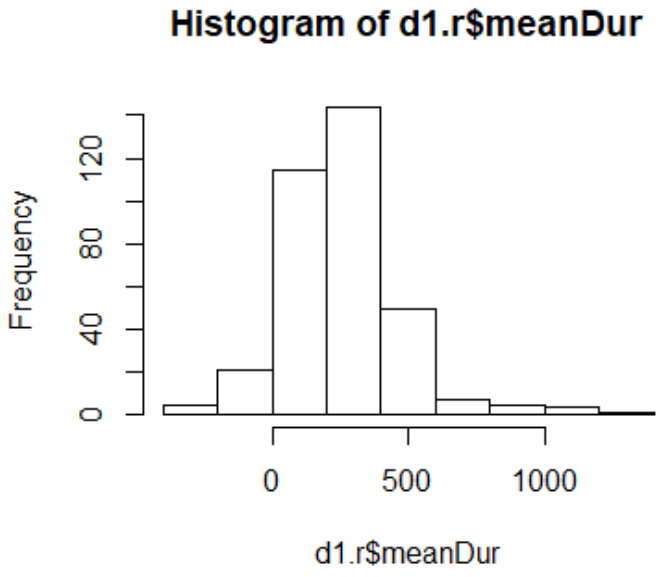
	file	meanDurFirst	meanDurAfter
1	sw3154.eaf	230.800000	244.8000000
2	sw3155.eaf	19.000000	152.7564103
3	sw3156.eaf	131.000000	-9.1111111
4	sw3159.eaf	165.500000	-28.2727273
5	sw3161.eaf	-351.500000	34.3125000
6	sw3168.eaf	31.666667	119.8250000
8	sw3171.eaf	-76.818182	-120.3035714
9	sw3174.eaf	378.200000	75.6923077
10	sw3182.eaf	329.615385	138.8157895

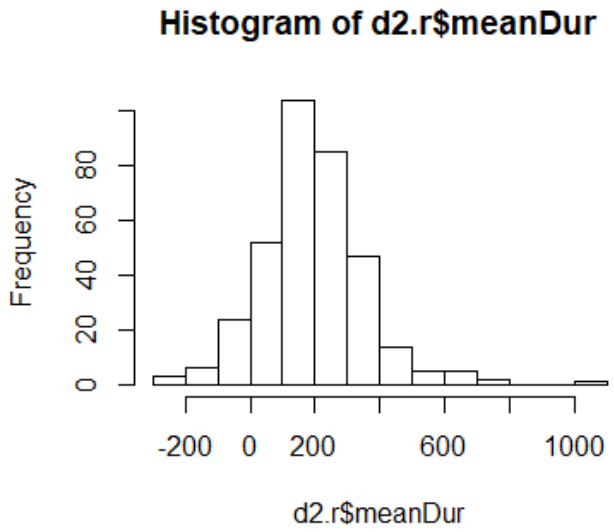


http://people.linguistics.mcgill.ca/~morgan/book/datasets-appendix.html#transitionsdata

Paired Samples





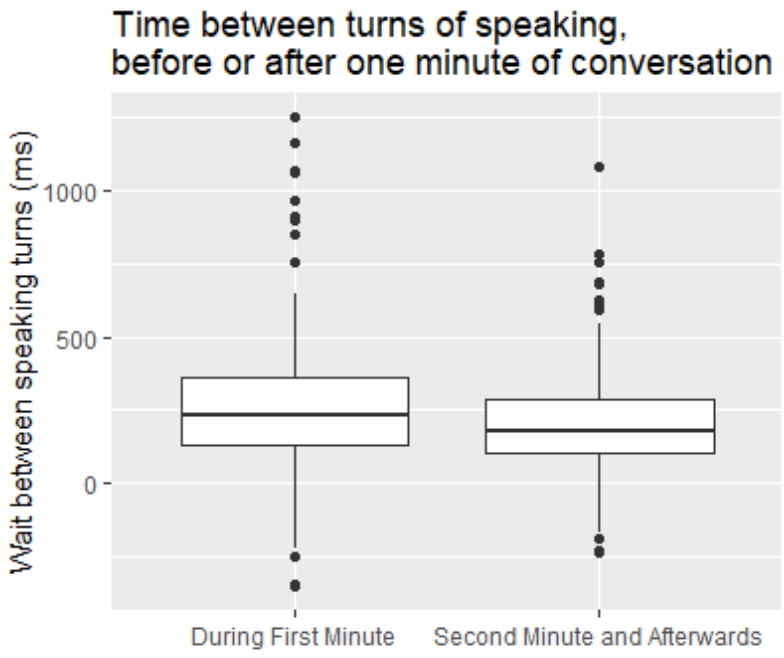


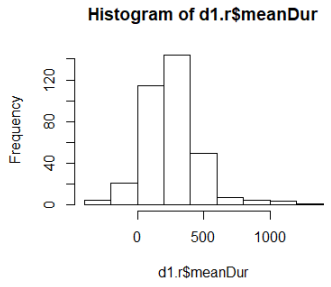
n = 347 n = 347

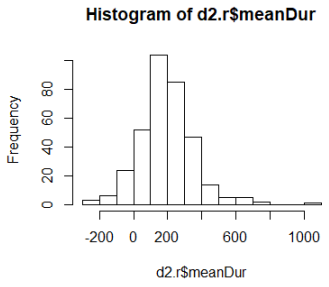
$\bar{x}_1$ = 256 ms $\bar{x}_2$ = 193 ms

http://people.linguistics.mcgill.ca/~morgan/qmld-book/datasets-appendix.html#transitionsdata

Paired Samples








```
t.test(d1.r$meanDur,d2.r$meanDur,
paired = TRUE)

        Paired t-test

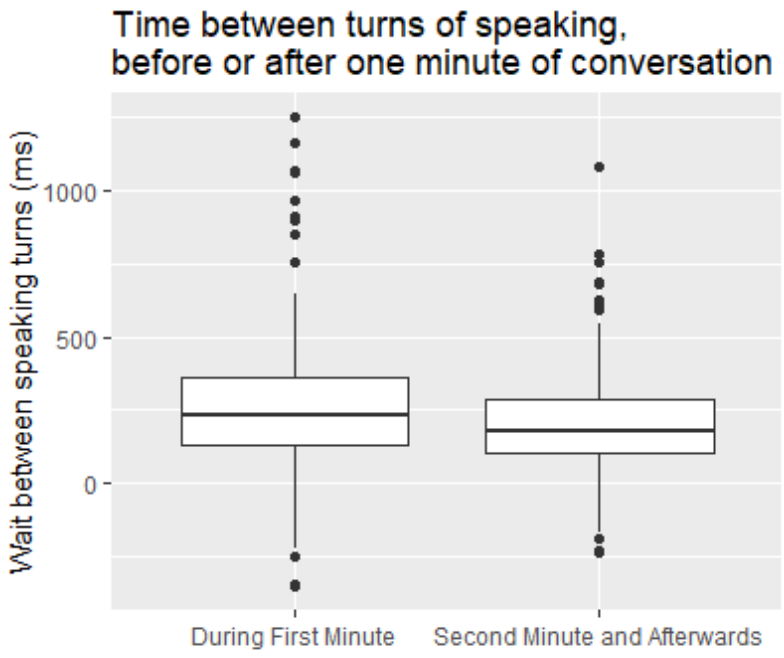
data:  d1.r$meanDur and d2.r$meanDur
t = 5.9893, df = 346, p-value = 5.283e-09

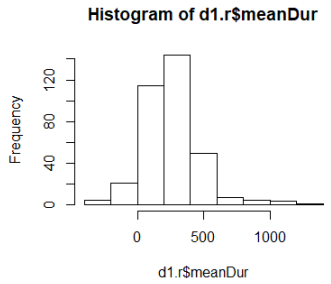
alternative hypothesis: true difference
in means is not equal to 0

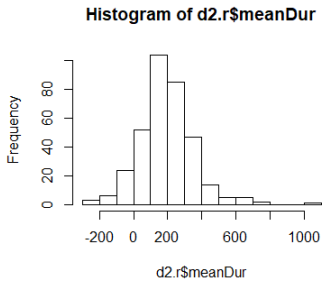
95 percent confidence interval:
 42.10084 83.27295

sample estimates:
mean of the differences
                 62.6869
```

Paired Samples








```
power.t.test(power=0.99929,
sig.level=0.05, delta=63, sd=195,
type="paired", alternative="two.sided")

     Paired t test power calculation

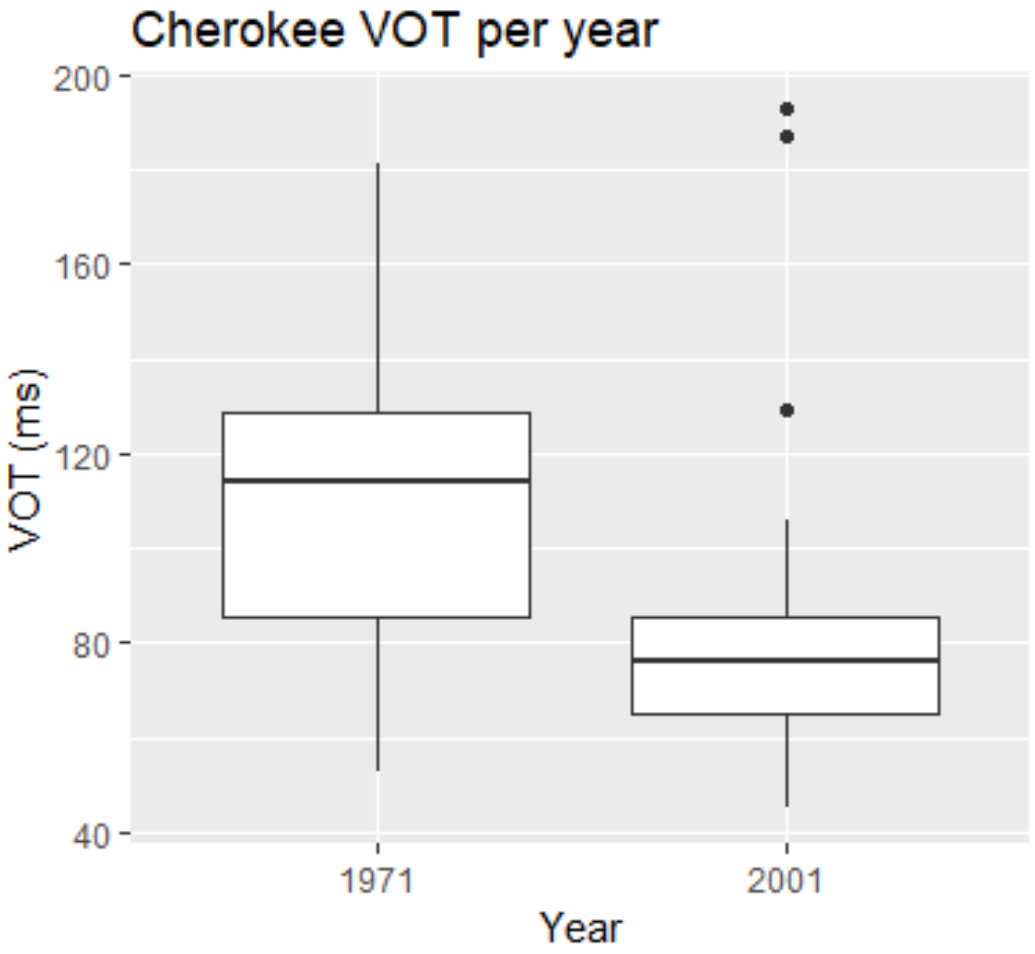
              n = 256.0831
          delta = 63
             sd = 195
      sig.level = 0.05
          power = 0.99929
    alternative = two.sided

NOTE: n is number of *pairs*, sd is
std.dev. of *differences* within pairs
```

How do you report these results: T-tests

How do we report this in a scientific paper?

How do you report these results: T-tests



SD_1 = 36 ms SD_2 = 36 ms

```
t.test(vot1971$VOT, vot2001$VOT)

      Welch Two Sample t-test

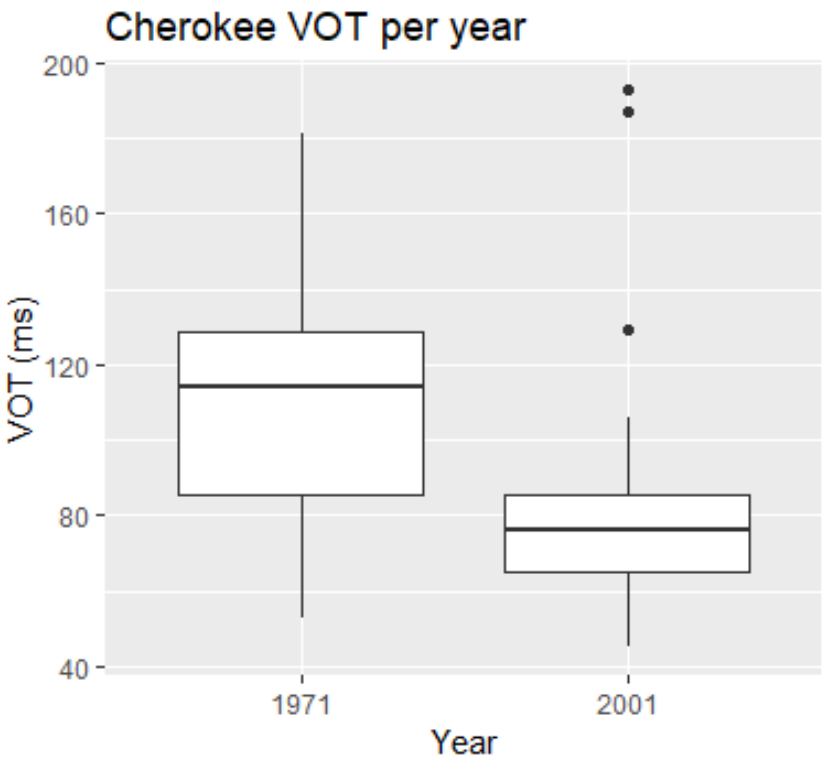
data:  vot1971$VOT and vot2001$VOT
t = 2.6137, df = 36.825, p-value = 0.0129

alternative hypothesis: true difference in
means is not equal to 0

95 percent confidence interval:
  6.480602 51.211706

sample estimates:
mean of x mean of y
113.50000  84.65385
```

How do you report these results: T-tests



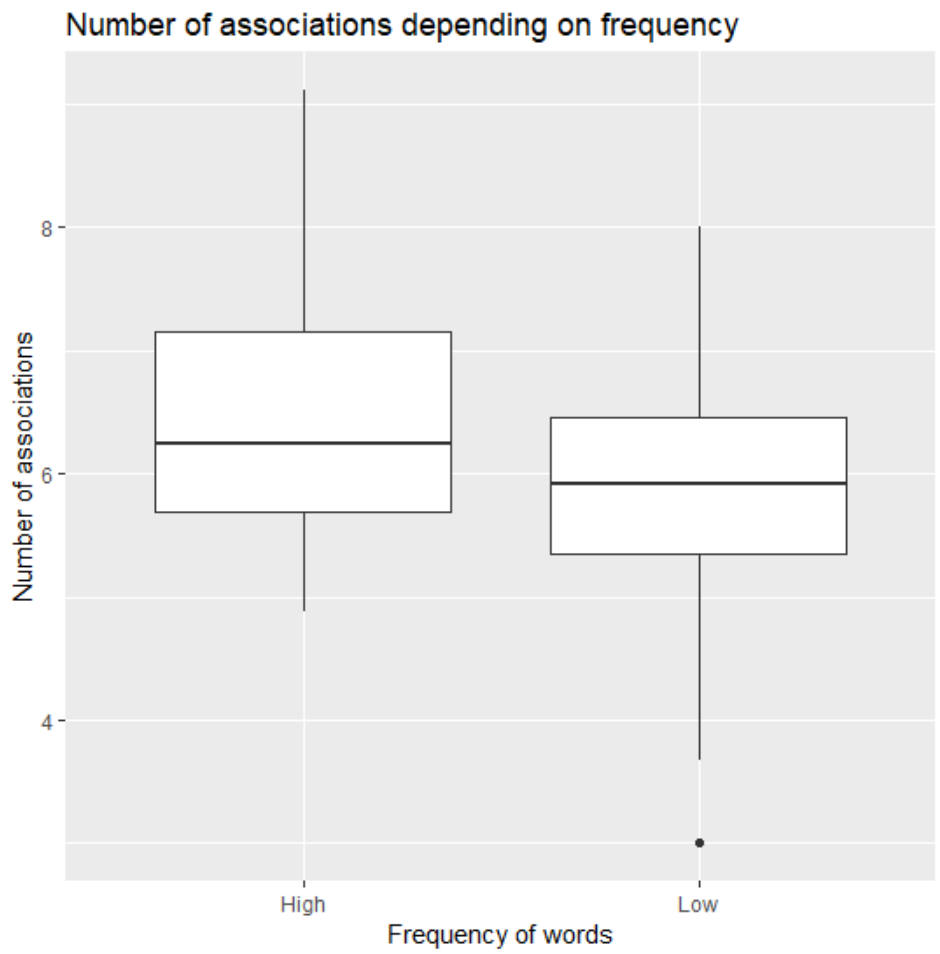

```
t.test(vot1971$VOT, vot2001$VOT)

        Welch Two Sample t-test
data:  vot1971$VOT and vot2001$VOT
t = 2.6137, df = 36.825, p-value = 0.0129
alternative hypothesis: true difference in means is
not equal to 0
95 percent confidence interval:
  6.480602 51.211706
sample estimates:
mean of x mean of y
113.50000  84.65385
```

SD_1 = 36 ms SD_2 = 36 ms

The VOT of Cherokee consonants differs significantly depending on the year ($t(36)=2.6$, $p < 0.05$). The VOT in 1971 was approximately 113 ms (SD=36 ms), significantly longer than the VOT in 2001, which was approximately 85 ms (SD=36 ms).

How do you report these results: T-tests



$n_1 = 50$ $n_2 = 51$

$\bar{x}_1 = 6.4$ $\bar{x}_2 = 5.9$

$SD_1 = 0.9$ $SD_2 = 1.03$

```
> t.test(pym_high$assoc, pym_low$assoc)

      Welch Two Sample t-test

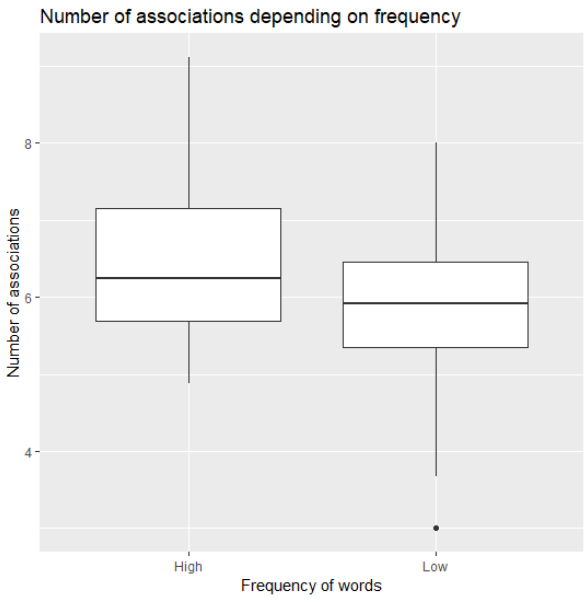
data:  pym_high$assoc and pym_low$assoc
t = 2.6717, df = 98.281, p-value = 0.008835

alternative hypothesis: true difference
in means is not equal to 0

95 percent confidence interval:
 0.1344293 0.9106687

sample estimates:
mean of x mean of y
 6.380000  5.857451
```

How do you report these results: T-tests

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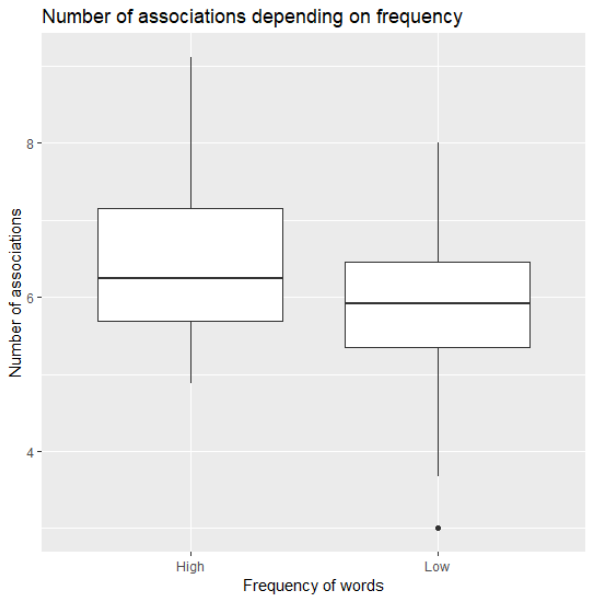
$\bar{x}_1 = 6.4$
$SD_1 = 0.9$

$n_2 = 51$
$\bar{x}_2 = 5.9$
$SD_2 = 1.03$

High frequency words are significantly different from low-frequency words (t(98)=2.7, p<0.01): High-frequency words have more associations (6.4, SD=0.9) than low frequency words (5.9, SD=1.03).

How do you report these results: T-tests




```
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        Welch Two Sample t-test
data:  pym_high$assoc and pym_low$assoc
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alternative hypothesis: true difference
in means is not equal to 0
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sample estimates:
mean of x mean of y
 6.380000  5.857451
```

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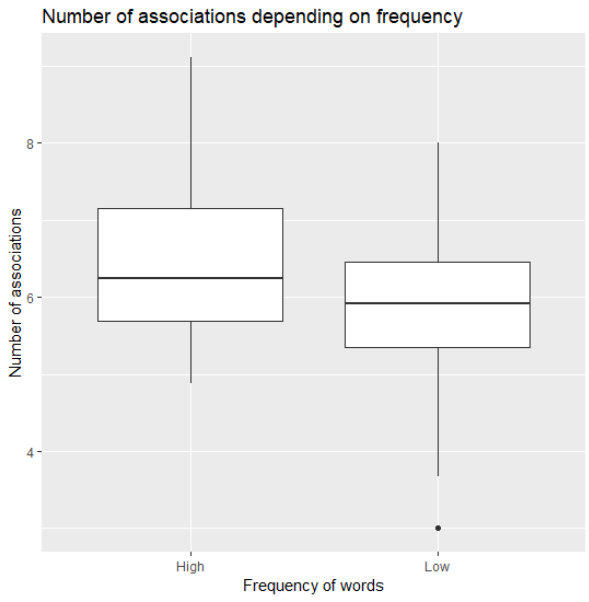
$\bar{x}_1 = 6.4$
$SD_1 = 0.9$

$n_2 = 51$
$\bar{x}_2 = 5.9$
$SD_2 = 1.03$

High frequency words are associated to more words (6.4, SD=0.9) than low frequency words (5.9, SD=1.03). This result is statistically significant (t(98)=2.7, p<0.01).

How do you report these results: T-tests

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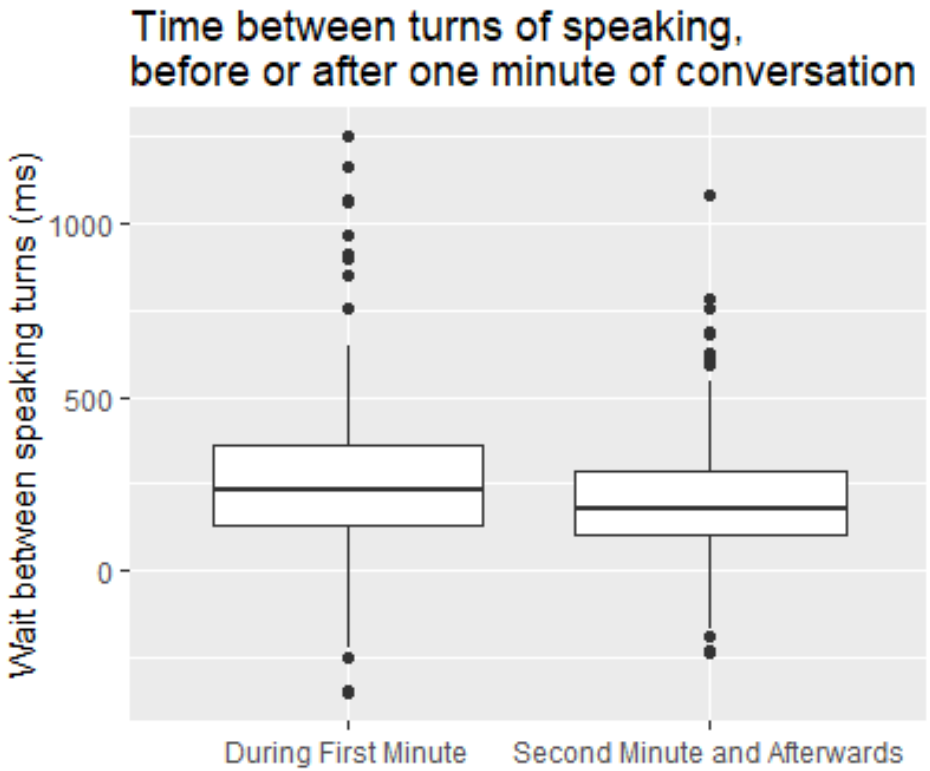
$\bar{x}_2 = 5.9$
$SD_2 = 1.03$

High frequency words are associated to more words (6.4, SD=0.9) than low frequency words (5.9, SD=1.03, t(98)=2.7, p<0.01).

How do you report these results: T-tests



$n_1 = 348$ $n_2 = 348$

$\bar{x}_1 = 256$ ms $\bar{x}_2 = 193$ ms

$SD_1 = 211$ ms $SD_2 = 164$ ms

```
t.test(d1.r$meanDur,d2.r$meanDur,
paired = TRUE)

      Paired t-test

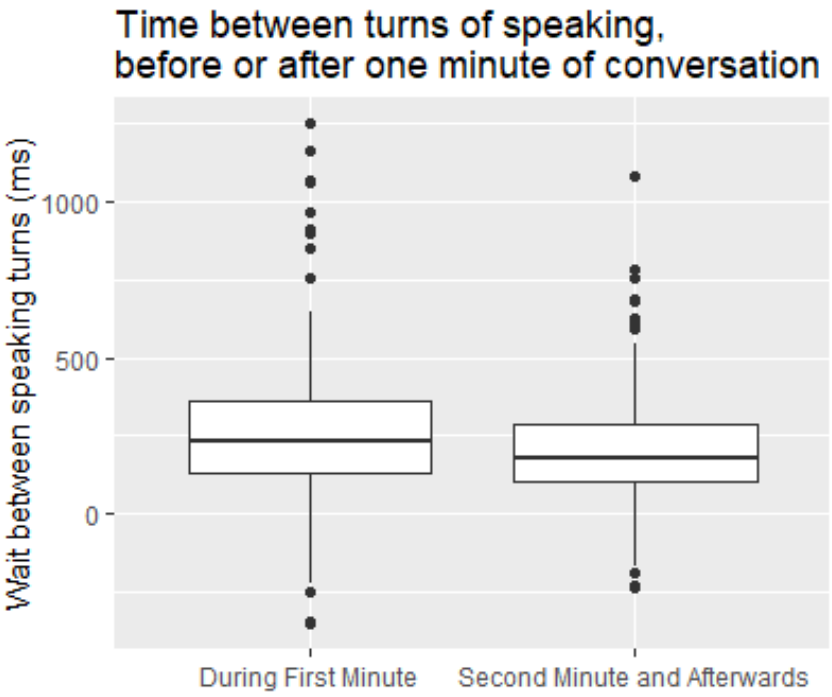
data:  d1.r$meanDur and d2.r$meanDur
t = 5.9893, df = 346, p-value = 5.283e-09

alternative hypothesis: true difference
in means is not equal to 0

95 percent confidence interval:
 42.10084 83.27295

sample estimates:
mean of the differences
                 62.6869
```

How do you report these results: T-tests




```
t.test(d1.r$meanDur,d2.r$meanDur,paired = TRUE)
        Paired t-test

data:  d1.r$meanDur and d2.r$meanDur
t = 5.9893, df = 346, p-value = 5.283e-09
alternative hypothesis: true difference in means is not
equal to 0

95 percent confidence interval:
 42.10084 83.27295
sample estimates:
mean of the differences
                 62.6869
```

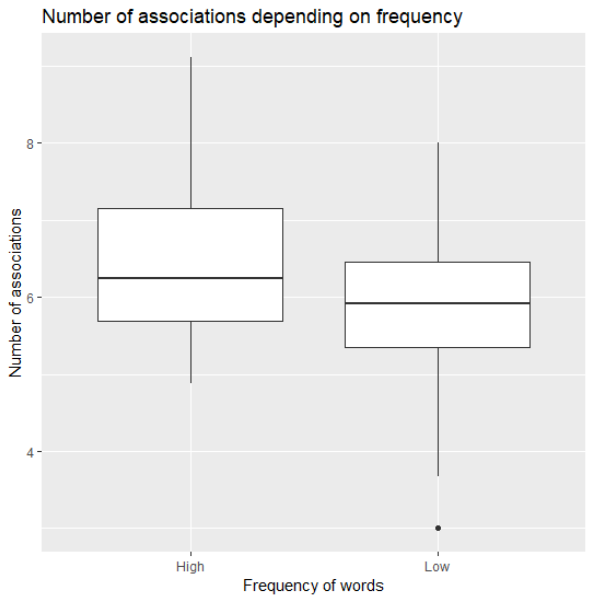
$n_1 = 348$ $n_2 = 348$

$\bar{x}_1 = 256$ ms $\bar{x}_2 = 193$ ms

$SD_1 = 211$ ms $SD_2 = 164$ ms

Time between turns is significantly longer in the first minute of the conversation than in subsequent minutes (paired t(346)=6.0, p<0.005). In the first minute the turns take approx. 256 ms (SD=211 ms), but after the first minute, the turns become shorter (mean=193 ms, SD=164 ms).

How do you report these results: T-tests



If, for example, p=0.20, we could simply say:

The number of associated words is not significantly different for high and low frequency words (t(98)=0.7, p=0.2).

The number of associated words is not significantly different for high and low frequency words (p=0.2).

Non-Parametric Tests

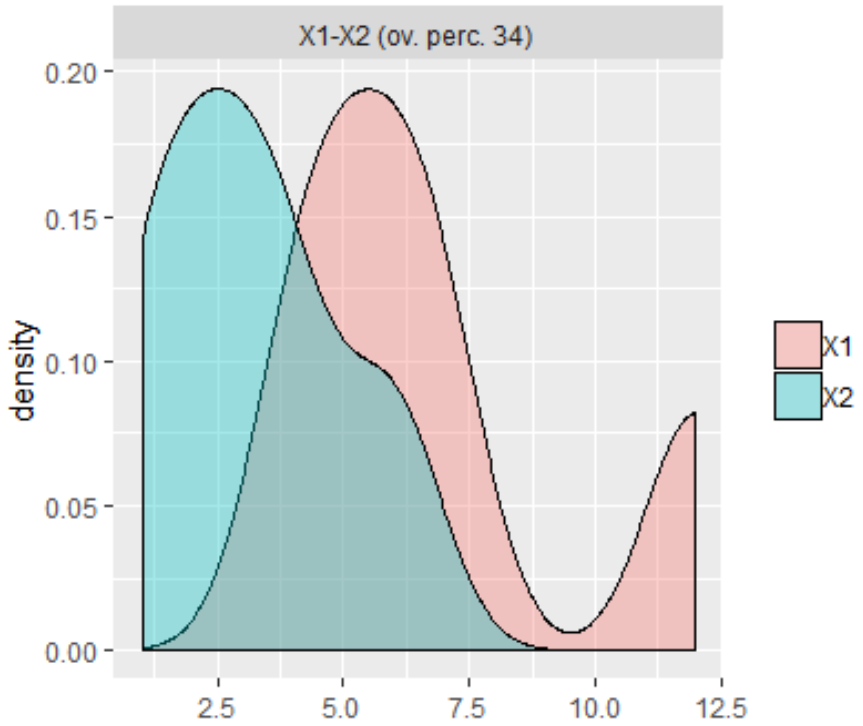
If your data isn't normal, you can use a **non-parametric test**.

They are less powerful, but they do not depend on distributions being normal.

A common substitute for the independent samples t-test is the **Mann-Whitney-U Test** (also, confusingly, called the Wilcoxon Rank Sum Test).

Non-Parametric Tests

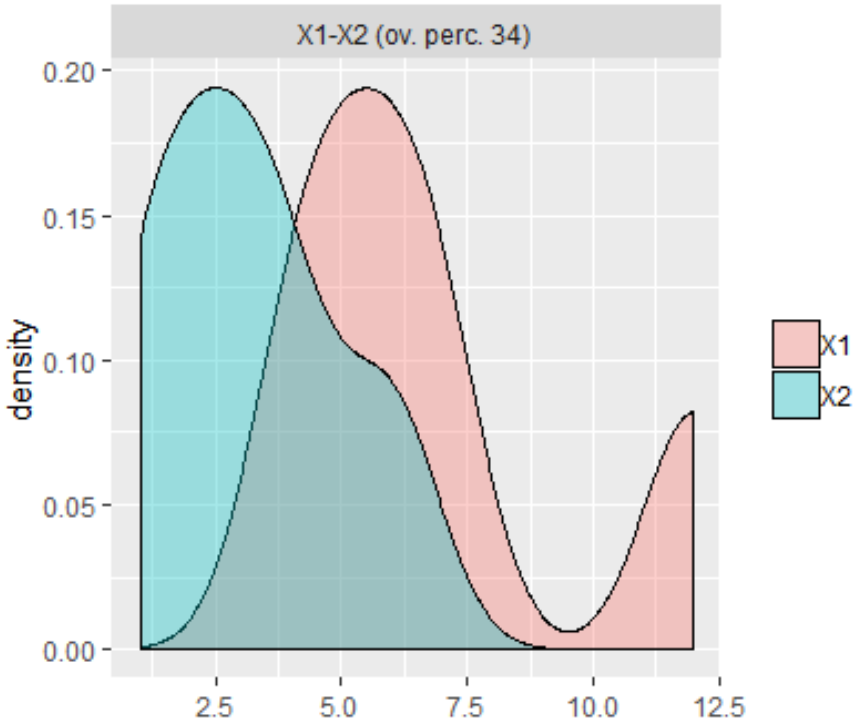
	Times that a character uses the word ‘hath’				
Novel1	7	5	6	4	12
Novel2	3	6	4	2	1



Non-Parametric Tests

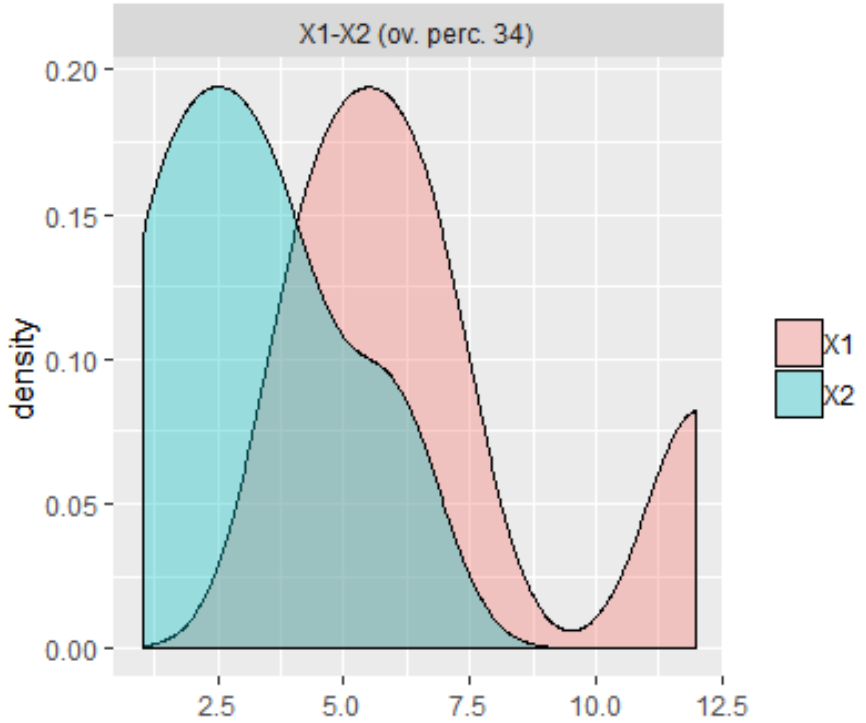
	Times that a character uses the word ‘hath’				
Novel1	7	5	6	4	12
Novel2	3	6	4	2	1

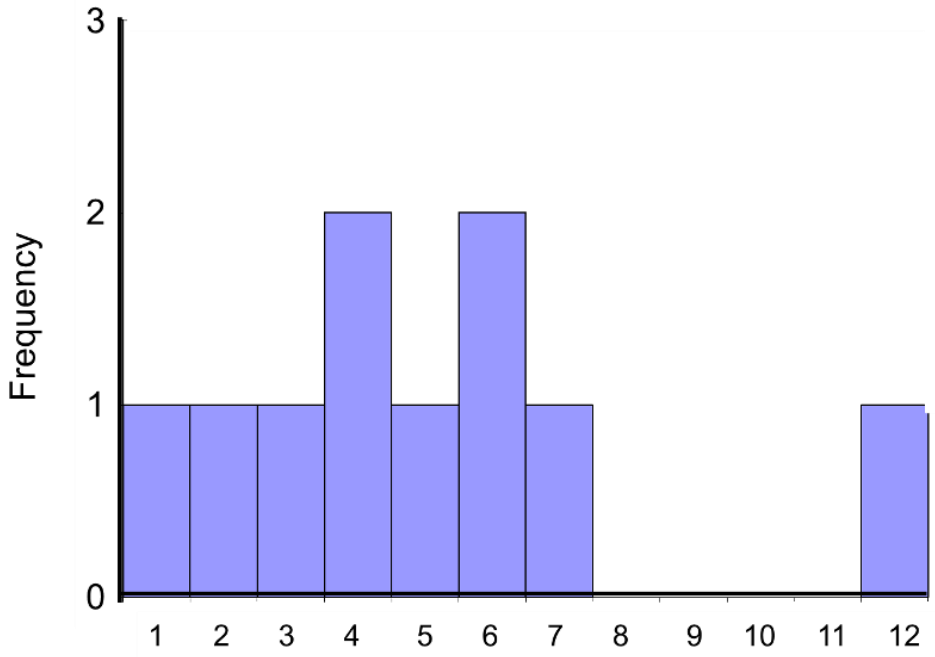
The only assumption: The distributions should have roughly the **same shape**.



Non-Parametric Tests

	Times that a character uses the word ‘hath’				
Novel1	7	5	6	4	12
Novel2	3	6	4	2	1





Times that characters use the word “hath”

Non-Parametric Tests

		Total Sample (from smallest to largest)		Ranks	
Novel1	**Novel2**	**Novel1**	**Novel2**	**Novel1**	**Novel2**
7	3		1		1
5	6		2		2
6	4		3		3
4	2	4	4	4.5	4.5
12	1	5		6	
		6	6	7.5	7.5
		7		9	
		12		10	
				37	18

Non-Parametric Tests

	Times that a characters uses the word 'hath'					Rank
Novel1	7	5	6	4	12	37
Novel2	3	6	4	2	1	18

$$W_1 = n_1 n_2 + \frac{n_1\,(n_1+1)}{2} - R_1 = (5)(5) + \frac{5(6)}{2} - 37 = 3$$

$$W_2 = n_1 n_2 + \frac{n_2\,(n_2+1)}{2} - R_2 = (5)(5) + \frac{5(6)}{2} - 18 = 22$$

$$Choosing\ the\ largest\ W_n, with\ \alpha = 0.05$$

$$W2\ > 10; therefore, this\ is\ significantly\ different$$

Non-Parametric Tests

	Times that a characters uses the word ‘hath’				
Novel1	7	5	6	4	12
Novel2	3	6	4	2	1

```
wilcox.test(numberHath ~ novel, data=hath,
alternative = c("greater"))

     Wilcoxon rank sum test with continuity correction

data:  numberHath by novel
W = 22, p-value = 0.02928

alternative hypothesis: true location shift is greater than 0
```

Non-Parametric Tests

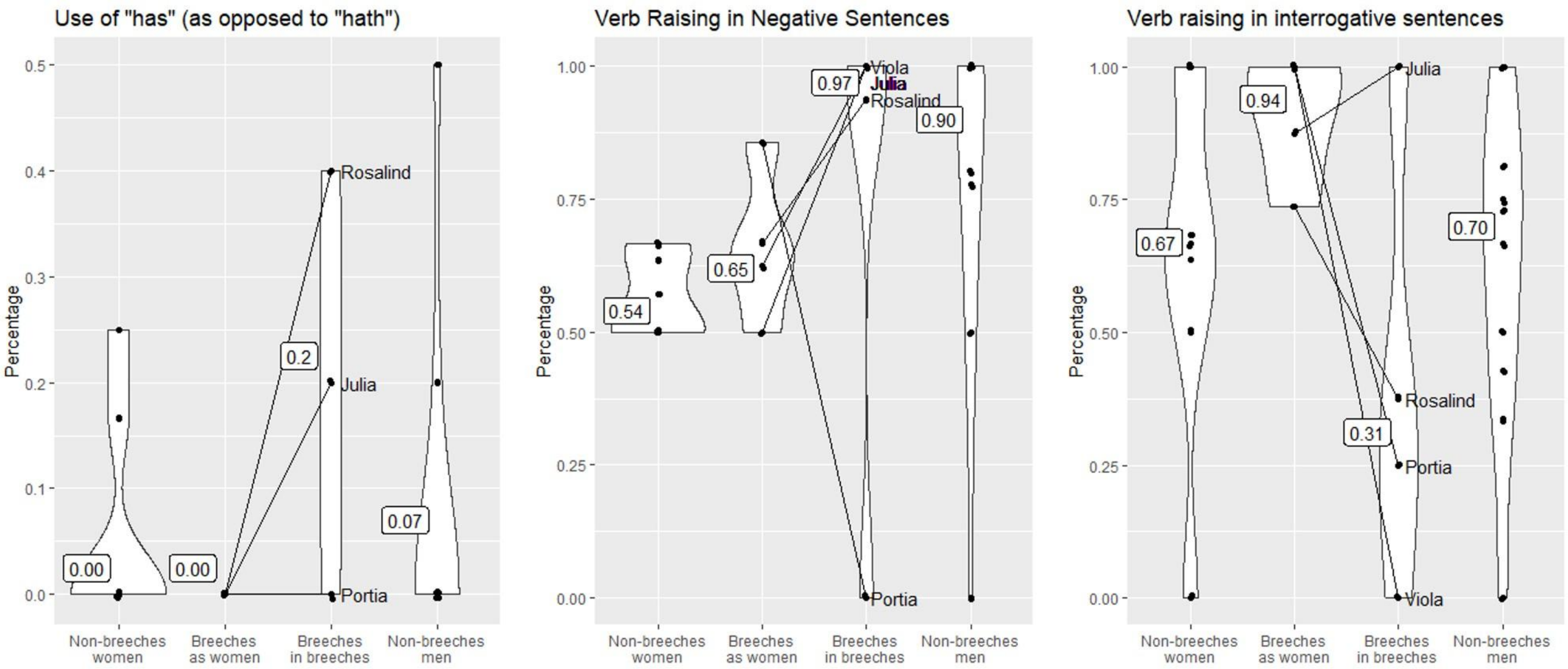
	Times that a character uses the word 'hath'					Median
Novel1	7	5	6	4	12	6
Novel2	3	6	4	2	1	3

```
wilcox.test(numberHath ~ novel, data=hath,
alternative = c("greater"))
      Wilcoxon rank sum test with continuity correction
data:  numberHath by novel
W = 22, p-value = 0.02928
alternative hypothesis: true location shift is greater than 0
```

A Mann-Whitney-Wilcoxon Test shows that characters in Novel1 use the verbal form hath *significantly more often (median=6, W=22, p=0.03) than characters in Novel2 (median=3).*

Non-Parametric Tests

https://www.degruyter.com/document/doi/10.1515/jhsl-2018-0022/html



The white squares have the median in them

Summary

If data is normal, a **t-test** is a powerful way to compare the means of two groups of data.

Even if the data is not normal, we still have options. The **Mann-Whitney-Wilcoxon** test lets us compare small groups of data that might not fulfill the assumptions of the t-test.

Next week, we will study how to compare **three** groups of data at the same time.